

Applied Stochastic Methods

Notes taken by: Kaichao Xu

Bases on the lectures by Prof Gavin Esler

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1 Stochastic Differential Equations(SDEs)

Stochastic Differential Equations are used to model CONTINUOUS-IN-TIME random processes.

Examples.

- (a) Stock prices in finance.
- (b) Motion of a dust particle in a turbulent flow.
- (c) Population of bacteria in a fluctuating environment.

1.1 Brownian motion

Definition 1.1. The Brownian process B_t is a stochastic process $t \geq 0$, which satisfies

1. Starts at the origin, $B_0 = 0$.
2. $B_t - B_s$ are normally distributed with mean zero and variance $t - s$ where WLOG we assume $t \geq s$,
$$B_t - B_s \sim N(0, t - s)$$
3. Non-overlapping increments are independent.

Remarks.

1. Recall $X \sim N(\mu, \sigma^2)$ has pdf

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

So $B_t - B_s$ has pdf

$$\lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(B_t - B_s \in [x, x + \delta x])}{\delta x} \equiv p(x, t, s) = \frac{1}{\sqrt{2\pi(t-s)}} \exp\left(-\frac{x^2}{2(t-s)}\right)$$

2. Also applies to $B_s = B_s - B_0$ so $B_t - B_s$ is independent of B_s and B_u for all $u < s$ which is a special case if the Markov process as defined below Markov process.
3. (Wiener) B_t is almost surely continuous everywhere and differentiable nowhere.
4. Notice that the pdf of B_t is

$$p(x, t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right)$$

and is a solution of the heat equation:

$$p_t = \frac{1}{2} p_{xx}$$

Definition 1.2. A stochastic process X_t satisfies the Markov process if the increment $X_t - X_s$ is independent at X_u for all $u < s$.

$$\mathbb{P}(X_t - X_s \in \text{Any set} | X_s, X_{u_1}, \dots, X_{u_n}) = \mathbb{P}(X_t - X_s \in \text{Any set} | X_s)$$

Let us think about B_t as the limit of a random walk

$$S_n = \sum_{i=1}^n \alpha X_i$$

where α is step-size and X_i Independently and Identically Distributed random walk. We will consider the case where $X_i \sim N(0, 1)$.

Lemma 1.3. Let X be a random variable with pdf $p_X(x)$ then $Y = \alpha X$ has pdf $p_Y(y) = \frac{1}{\alpha} p_X(\frac{y}{\alpha})$.

Proof.

$$\begin{aligned} p_Y(y) &= \lim_{\delta y \rightarrow 0} \frac{\mathbb{P}(Y \in [y, y + \delta y])}{\delta y} = \lim_{\delta y \rightarrow 0} \frac{\mathbb{P}(\alpha X \in [y, y + \delta y])}{\delta y} \\ &\Rightarrow p_Y(y) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(X \in [\frac{y}{\alpha}, \frac{y}{\alpha} + \delta x])}{\alpha \delta x} = \frac{1}{\alpha} p_X\left(\frac{y}{\alpha}\right) \end{aligned}$$

□

Lemma 1.4. Let $Y_1 \sim N(0, \alpha_1^2)$ and $Y_2 \sim N(0, \alpha_2^2)$ Then $Z = Y_1 + Y_2 \sim N(0, \alpha_1^2 + \alpha_2^2)$.

Proof. We can write the pdf of Z as an integral

$$p_Z(z) = \int_{-\infty}^{\infty} p_{X_1}(y) p_{X_2}(z - y) dy$$

and $p_Z = p_{Y_1} * p_{Y_2}$ and if we define the Fourier transform to be

$$\hat{p}_Z(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p_{Y_1}(y) e^{-iuy} dy$$

then

$$\begin{aligned} \mathcal{F}\left[\frac{1}{\sqrt{2\pi}\alpha_1} \exp\left(-\frac{y^2}{2\alpha_1^2}\right)\right] &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2\alpha_1^2}{2}\right) \\ \hat{p}_Z(u) &= \sqrt{2\pi} \hat{p}_{Y_1}(u) \hat{p}_{Y_2}(u) \\ \Rightarrow \hat{p}_Z(u) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\alpha_1^2 + \alpha_2^2)u^2}{2}\right) \end{aligned}$$

which is the Fourier transform of the Gaussian $N(0, \alpha_1^2 + \alpha_2^2)$. $\square$

Proposition 1.5. (Law of addition of Gaussian random variables) Consider $X_i \sim N(0, 1)$ (i.i.d.) and $\alpha_i \in \mathbb{R}$ and $S_n^* = \sum_{i=1}^n \alpha_i X_i$ then $S_n^* \sim N(0, \sum_{i=1}^n \alpha_i^2)$.

Proof. X_i has pdf $p_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ say $Y_i = \alpha_i X_i$ then

$$p_{Y_i}(y) = \frac{1}{\sqrt{2\pi}\alpha_i} e^{-y^2/2\alpha_i^2}$$

so $Y_i \sim N(0, \alpha_i^2)$. Apply two lemmas above we immediately have

$$S_n^* \sim N(0, \sum_{i=1}^n \alpha_i^2)$$

$\square$

In our original sum

$$S_n = \sum_{i=1}^n \alpha X_i$$

then $S_n \sim N(0, n\alpha^2)$. Introduce a "discrete time variable" $t_n = nh$ where $h = \text{time-step}$.

Definition 1.6. Define a discrete "Brownian" process

$$\tilde{B}(t_n) = S_n$$

Note that $S_n \sim N(0, \frac{t_n \alpha^2}{h})$. Choose $\alpha = \sqrt{h}$ then $\tilde{B}(t_n) \sim N(0, t_n)$.

Note that $\tilde{B}(0) = 0$. Now consider $\tilde{B}(t_n) - \tilde{B}(s_n)$ then

$$\tilde{B}(t_n) - \tilde{B}(s_n) = \sum_{i=s_n/h+1}^{t_n/h} \sqrt{h} X_i \sim N(0, t_n - s_n)$$

to check non-overlapping increments independent:

$$\begin{aligned} \tilde{B}(t_{n_4}) - \tilde{B}(t_{n_3}) &= \sum_{i=t_{n_3}/h+1}^{t_{n_4}/h} \sqrt{h} X_i \\ \tilde{B}(t_{n_2}) - \tilde{B}(t_{n_1}) &= \sum_{i=t_{n_1}/h+1}^{t_{n_2}/h} \sqrt{h} X_i \end{aligned}$$

different X_i s so sums are independent.

Theorem 1.7. (Central Limit Theorem) The X_i 's in the above discussion did not need to be Gaussian. Let X_i be i.i.d. random variables with mean μ and finite variance σ^2 . Let $S_n = \sum_{i=1}^n X_i$ then

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \rightarrow N(0, 1)$$

in distribution. From law of addition if $\mu = 0$ then

$$S_n \sim N(0, n\sigma^2)$$

so any X_i will be for definition of B_t provided that $\sigma^2 = h$.

Remark. It is possible to construct other processes (Levy Flights) starting from X_i with infinite variance.

Properties of B_t :

(a)

$$\begin{aligned}\mathbb{E}(f(B_t)) &= \int_{-\infty}^{\infty} f(x)p(x,t) dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} f(x)e^{-\frac{x^2}{2t}} dx\end{aligned}$$

(b) Say $0 \leq s < t$ then

$$\begin{aligned}\mathbb{E}(B_s B_t) &= \mathbb{E}(B_s(B_t - B_s + B_s)) \\ &= \mathbb{E}(B_s(B_t - B_s)) + \mathbb{E}(B_s^2) = s\end{aligned}$$

In general,

$$E(B_s B_t) = \min\{s, t\}$$

(c) Let B_t be a Brownian process, define

$$W_\tau = \frac{1}{\sqrt{c}} B_{c\tau}$$

then W_τ is a Brownian process with the new time variable $\tau = \frac{t}{c}$.

(d) Let B_t and W_t be two independent Brownian processes, define $Q_t = aB_t + \sqrt{1-a^2}W_t$ where $0 \leq a \leq 1$. Then Q_t is a Brownian process.

1.2 Stochastic Integration

SDEs are (most formally) expressed as Integral equations. We need to think about integrals. In particular, we need to consider

$$\begin{aligned}I(t) &= \int_0^t f(B_s, s) dB_s \\ J(t) &= \int_0^t f(B_s, s) ds\end{aligned}$$

Definition 1.8. $I(t)$ requires precise definition. Two possibilities are:

(a) Itô

$$I_I(t) = \lim_{h \rightarrow 0} \sum_{i=1}^n f(B_{t_i}, t_i)(B_{t_{i+1}} - B_{t_i})$$

(b) Stratonovich

$$I_S(t) = \lim_{h \rightarrow 0} \sum_{i=1}^n f(B_{t_{i+1/2}}, t_{i+1/2})(B_{t_{i+1}} - B_{t_i})$$

We need to consider integrals with respect to dB_t^2 e.g.

$$\int_0^t f(B_s, s)(dB_s)^2 = \lim_{h \rightarrow 0} \sum_i^n f(B_{t_i}, t_i)(B_{t_{i+1}} - B_{t_i})^2$$

It turns out that, for Itô,

$$\int_0^t f(B_s, s)(dB_s)^2 = \int_0^t f(B_s, s) ds$$

almost surely. This allows us to define the increments rules of Itô calculus

$$dB_t^2 = dt, \quad dt dB_t = dt^2 = 0$$

Definition 1.9. (Non-anticipating property)

$$\begin{aligned}\mathbb{E}\left(\int_0^t f(B_s, s) dB_s\right) &= \lim_{h \rightarrow 0} \mathbb{E}\left(\sum_i f(B_{t_i}, t_i)(B_{t_{i+1}} - B_{t_i})\right) \\ &= \lim_{h \rightarrow 0} \mathbb{E}(f(B_{t_i}, t_i))\mathbb{E}(B_{t_{i+1}} - B_{t_i}) = 0\end{aligned}$$

Let $f(x, t)$ be smooth function of two variables. Define $f(t) = f(B(t), t)$ the value of $f(x, t)$ along a path $x = B(t)$. Then

$$\frac{df}{dt} = \frac{\partial f}{\partial t}(B(t), t) + \frac{dB_t}{dt} \frac{\partial f}{\partial x}(B(t), t)$$

so

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dB_t$$

Definition 1.10. (Itô formula) By Taylor expansion,

$$f(B_{t+dt}, t+dt) = f(B_t, t) + \frac{\partial f}{\partial t}(B_t, t)dt + \frac{\partial f}{\partial x}(B_t, t)dB_t$$

$$+ \frac{1}{2} \left(\frac{\partial^2 f}{\partial t^2} (B_t, t) dt^2 + 2 \frac{\partial^2 f}{\partial x \partial t} (B_t, t) dB_t dt + \frac{\partial^2 f}{\partial x^2} (B_t, t) (dB_t)^2 \right)$$

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dB_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} dt$$

Example. What is $\int_0^t B_s dB_s$?

Try Itô on $f(B_t) = B_t^2$ then

then

$$d(B_t^2) = 2B_t dB_t + dt$$

$$B_t^2 = \int_0^t 2B_s dB_s + \int_0^t ds$$

$$B_t^2 - t = 2 \int_0^t B_s dB_s$$

$$\int_0^t B_s dB_s = \frac{B_t^2 - t}{2}$$

Example. Simplify $\int_0^t B_s^2 dB_s$.

Consider $d(B_t^3)$ then

$$d(B_t^3) = 3B_t^2 dB_t + 3B_t dt$$

take the integral from 0 to t to give

$$B_t^3 = \int_0^t 3B_s^2 dB_s + \int_0^t B_s ds$$

which gives

$$\int_0^t B_s^2 dB_s = \frac{1}{3} B_t^3 - \int_0^t B_s ds$$

Properties of Stochastic integrals

(a) Non-anticipating property

$$\mathbb{E} \left(\int_0^t f(B_s, s) dB_s \right) = 0$$

(b)

$$\mathbb{E} \left(\left(\int_0^t f(B_s, s) dB_s \right)^2 \right) = \mathbb{E} \left(\int_0^t (f(B_s, s))^2 ds \right)$$

(c)

$$\int_0^t f(s) dB_s \sim N(0, \int_0^t f(s)^2 ds)$$

Let $B_t^{(1)}, B_t^{(2)}, \dots, B_t^{(n)}$ be independent Brownian processes then

(1) Increments rules. $dt^2 = dt dB_t^{(i)} = 0$ and $dB_t^{(i)} dB_t^{(j)} = \delta_{ij} dt$.

(2)

$$f(B_{t+dt}^{(1)}, \dots, B_{t+dt}^{(n)}, t+dt) =$$

$$f(B_t^{(1)}, \dots, B_t^{(n)}, t) + \frac{\partial f}{\partial t} dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i} dB_t^{(i)} + \frac{1}{2} \left(\frac{\partial^2 f}{\partial t^2} dt^2 + 2 \sum_{i=1}^n \frac{\partial f}{\partial x_i \partial t} dB_t^{(i)} dt + \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j} dB_t^{(i)} dB_t^{(j)} \right)$$

Let $\underline{x} = (x_1, \dots, x_n)$ and $\underline{B}_t = (B_t^{(1)}, \dots, B_t^{(n)})$ then

$$df = f_t dt + \nabla f \cdot d\underline{B}_t + \frac{1}{2} \nabla^2 f dt$$

$$\implies df = (f_t + \frac{1}{2} \nabla^2 f) dt + \nabla f \cdot d\underline{B}_t$$

which is the Itô formula for n dimension.

1.3 Stochastic differential equations

Idea: Introduce a new class of stochastic processes X_t generated by integral equations. The general one dimensional SDE:

$$dX_t = a(X_t, t) dt + b(X_t, t) dB_t$$

where $a(X_t, t)$ is called the drift and $b(X_t, t)$ is the noise function and b^2 is diffusion. Shorthand for

$$X_t - X_0 = \int_0^t a(X_s, s) ds + \int_0^t b(X_s, s) dB_s$$

Definition 1.11. If a and b have no explicit time independence, i.e.,

$$dX_t = a(X_t) dt + b(X_t) dB_t$$

the equation is said to be autonomous.

Definition 1.12. (Itô to SDEs)

$$\begin{aligned} f(X_{t+dt}, t+dt) &= f(X_t, t) + \frac{\partial f}{\partial x} dX_t + \frac{\partial f}{\partial t} dt + \frac{1}{2} \left(\frac{\partial^2 f}{\partial t^2} dt^2 + 2 \frac{\partial^2 f}{\partial t \partial x} dX_t dt + \frac{\partial^2 f}{\partial x^2} dX_t^2 \right) \\ &= f(X_t, t) + \frac{\partial f}{\partial x} dX_t + \frac{\partial f}{\partial t} dt + \frac{1}{2} \left(\frac{\partial^2 f}{\partial x^2} dX_t^2 \right) \end{aligned}$$

thus,

$$df = (f_t + af_x + \frac{1}{2}b^2 f_{xx})dt + bf_x dB_t$$

Definition 1.13. (The generator of an SDE) This is a differential operator defined on a wide class of processes X_t . The operator $\mathcal{L}$ acts on smooth functions $g(x)$ according to

$$\mathcal{L}g = \lim_{s \rightarrow 0} \frac{\mathbb{E}(g(X_{t+s}|x_t = x)) - g(x)}{s}$$

then

$$\begin{aligned} \mathcal{L}g &= \lim_{s \rightarrow 0} \frac{\mathbb{E}(g(X_{t+s} - g(X_t)|x_t = x))}{s} \\ &= \lim_{dt \rightarrow 0} \frac{\mathbb{E}(g(X_{t+dt} - g(X_t)|x_t = x))}{dt} \\ &= \lim_{dt \rightarrow 0} \frac{\mathbb{E}(dg|X_t = x)}{dt} \\ &= \lim_{dt \rightarrow 0} \frac{\mathbb{E}(b(X_t, t)g_x(X_t)dB_t + a(X_t, t)g_x(X_t)dt + \frac{1}{2}b(X_t, t)g_{xx}(X_t)dt|X_t = x)}{dt} \\ &= \lim_{dt \rightarrow 0} \frac{\mathbb{E}(a(X_t, t)g_x(X_t)dt + \frac{1}{2}b(X_t, t)g_{xx}(X_t)dt|X_t = x)}{dt} \\ &= a(x, t)g_x + \frac{1}{2}b(x, t)g_{xx} \end{aligned}$$

thus,

$$\mathcal{L} \equiv a(x, t) \frac{d}{dx} + \frac{1}{2}b(x, t)^2 \frac{d^2}{dx^2}$$

is the generator.

1.4 Some SDEs and solution methods

Example.

$$dX_t = \frac{n(n-1)}{2} X_t^{\frac{n-2}{n}} dt + nX_t^{\frac{n-1}{n}} dB_t$$

subjected to the initial condition $X_0 = 0$. Show that $X_t = B_t^n$ is a solution.

Proof. By Itô calculus,

$$\begin{aligned} d(B_t^n) &= nB_t^{n-1}dt + \frac{n(n-1)}{2}B_t^{n-2}dB_t \\ &= n(B_t^n)^{\frac{n-1}{n}}dt + \frac{n(n-1)}{2}(B_t^n)^{\frac{n-2}{n}}dB_t \\ &= \frac{n(n-1)}{2}X_t^{\frac{n-2}{n}}dt + nX_t^{\frac{n-1}{n}}dB_t \end{aligned}$$

□

Consider

$$dX_t = a(X_t)dt + B(X_t)dB_t$$

when does the SDE gives solutions $X_t = f(B_t)$? By Itô calculus,

$$dX_t = df = f'(B_t)dt + \frac{1}{2}f''(B_t)dB_t$$

then

$$\begin{aligned} f'(B_t) &= b(f(B_t)) & \frac{1}{2}f''(B_t) &= a(f(B_t)) \\ f''(x) &= b'(f(x))f'(x) &= b'(f(x))b(f(x)) \\ &\implies 2a(f(x)) &= b'(f(x))b(f(x)) \end{aligned}$$

Re-label $f(x) = y$ then $a = \frac{1}{2}b'b$ known as the reducibility condition. For reducible equation, just need to solve first equation.

$$f'(x) = b(f(x))$$

use $y = f(x)$ then $y' = b(y)$ so

$$x + c = \int^y \frac{ds}{b(s)} = h(y)$$

solution is $y = f(x) = h^{-1}(x + c)$ where $h(y) = \int^y \frac{ds}{b(s)}$. Use the initial condition $X_0 = 0$ to get $c = h(X_0)$

$$\implies X_t = h^{-1}(B_t + h(X_0))$$

which is the general solution of the reducible equation.

Exercise. Show that first example is irreducible and use this method to solve it.

Example. Geometric Brownian Motion (GBM)

$$dX_t = \mu X_t dt + \sigma X_t dB_t$$

subjected to the initial condition $X_0 = x_0$.

Apply Itô to $Y_t = \log(X_t)$ then

$$\begin{aligned} d(\log X_t) &= \frac{dX_t}{X_t} - \frac{(dX_t)^2}{2X_t^2} \\ &= \mu dt + \sigma dB_t - \frac{\sigma^2 X_t^2 dt}{2X_t^2} \\ &= \mu dt + \sigma dB_t - \frac{1}{2} \sigma^2 dt \\ &= \left(\mu - \frac{1}{2} \sigma^2\right) dt + \sigma dB_t \end{aligned}$$

Integrate yields

$$\begin{aligned} \log X_t - \log X_0 &= \mu t + \sigma B_t - \frac{1}{2} \sigma^2 t \\ X_t &= x_0 \exp\left(\left(\mu - \frac{1}{2} \sigma^2\right)t + \sigma B_t\right) \\ \mathbb{E}(X_t) &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} x_0 \exp\left(\left(\mu - \frac{1}{2} \sigma^2\right)t + \sigma B_t\right) e^{-\frac{x^2}{2t}} dt \\ \mathbb{E}(X_t) &= x_0 e^{\mu t} \end{aligned}$$

which is the mean return. The medium return is

$$x_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t\right)$$

Example. (Ornstein-Uhlenbeck (O-U) process)

$$dX_t = -\frac{X_t}{\tau} dt + \left(\frac{2\sigma^2}{\tau}\right)^{\frac{1}{2}} dB_t$$

where σ is the standard deviation and τ is the decorrelation timescale.

Use the integrating factor $e^{t/\tau}$ and apply the Itô calculus to $d(X_t e^{t/\tau})$ then

$$\begin{aligned} d(X_t e^{t/\tau}) &= \frac{X_t}{\tau} e^{t/\tau} dt + e^{t/\tau} dX_t \\ \implies d(X_t e^{t/\tau}) &= \frac{X_t}{\tau} e^{t/\tau} dt - \frac{X_t}{\tau} e^{t/\tau} dt + e^{t/\tau} \left(\frac{2\sigma^2}{\tau}\right)^{\frac{1}{2}} dB_t \\ \implies X_t e^{t/\tau} - X_0 e^{0/\tau} &= \left(\frac{2\sigma^2}{\tau}\right)^{\frac{1}{2}} \int_0^t e^{s/\tau} dB_s \\ \implies X_t &= X_0 e^{-t/\tau} + e^{-t/\tau} \left(\frac{2\sigma^2}{\tau}\right)^{\frac{1}{2}} \int_0^t e^{s/\tau} dB_s \end{aligned}$$

Alternative method: $\mathbb{E}(X_t)$: say $y = \mathbb{E}(X_t)$ then $dy = -\frac{y}{\tau} dt$ with $y(0) = x_0$ then $y(t) = x_0 e^{-t/\tau}$. And

$$d(X_t^2) = 2X_t dX_t + (dX_t)^2 = 2X_t \left(-\frac{X_t}{\tau} dt + \left(\frac{2\sigma^2}{\tau}\right)^{\frac{1}{2}} dB_t\right) + \frac{2\sigma^2}{\tau} dt$$

then let $z = \mathbb{E}(X_t^2)$ where $z(0) = x_0^2$

$$dz = -\frac{2z}{\tau} dt + \frac{2\sigma^2}{\tau} dt$$

z is the solution of the differential equation

$$\frac{dz}{dt} + \frac{2z}{\tau} = \frac{2\sigma^2}{\tau}$$

Example. (General linear SDEs)

$$dX_t = (a(t)X_t + b(t))dt + (c(t)X_t + d(t))dB_t$$

The homogeneous equation looks like

$$\begin{aligned} dY_t &= a(t)Y_t dt + c(t)Y_t dB_t \\ d(\log Y_t) &= \frac{dY_t}{Y_t} - \frac{1}{2Y_t^2} (dY_t)^2 \\ &= a(t)dt + c(t)dB_t - \frac{c(t)^2}{2} dt \end{aligned}$$

Set $Y_0 = 1$ then

$$\begin{aligned}\log Y_t - \log Y_0 &= \int_0^t \left(a(s) - \frac{1}{2}c(s)^2 \right) ds + \int_0^t c(s) dB_s \\ Y_t &= \exp \left(\int_0^t \left(a(s) - \frac{1}{2}c(s)^2 \right) ds + \int_0^t c(s) dB_s \right)\end{aligned}$$

Use integrating factor of Y_t^{-1} then

$$\begin{aligned}d\left(\frac{X_t}{Y_t}\right) &= \frac{dX_t}{Y_t} - \frac{X_t dY_t}{Y_t^2} + \frac{X_t (dY_t)^2}{Y_t^3} - \frac{dX_t dY_t}{Y_t^2} \\ &= \frac{(aX_t + b)dt + (cX_t + d)dB_t}{Y_t} - \frac{X_t(aY_t dt + cY_t dB_t)}{Y_t^2} - \frac{(cX_t + d)cY_t dt}{Y_t^2} + \frac{c^2 Y_t^2 X_t dt}{Y_t^3} \\ &= \frac{(b(t) - c(t)d(t))dt + d(t)dB_t}{Y_t}\end{aligned}$$

Integrating yields

$$\frac{X_t}{Y_t} - \frac{x_0}{1} = \int_0^t \frac{(b(s) - c(s)d(s))}{Y_s} ds + \int_0^t \frac{d(s)}{Y_s} dB_s$$

Thus,

$$X_t = Y_t \left(x_0 + \int_0^t \frac{(b(s) - c(s)d(s))}{Y_s} ds + \int_0^t \frac{d(s)}{Y_s} dB_s \right)$$

where Y_t is known.

Example. (Noise removal transformation) Idea: Use integration factor with a wide class of equations to remove the noise term. Consider

$$dX_t = f(X_t, t)dt + c(t)X_t dB_t$$

Try to remove the noise term.

First solve $dY_t = c(t)Y_t dB_t$, note that

$$d(\log Y_t) = \frac{dY_t}{Y_t} - \frac{2(dY_t)^2}{Y_t^2} = c(t)dB_t - \frac{1}{2}c(t)^2 dt$$

then

$$Y_t = \exp \left(\int_0^t c(s) dB_s - \frac{1}{2} \int_0^t c(s)^2 ds \right)$$

Use the integrating factor Y_t^{-1} then

$$\begin{aligned}d\left(\frac{X_t}{Y_t}\right) &= \frac{dX_t}{Y_t} - \frac{X_t dY_t}{Y_t^2} - \frac{dX_t dY_t}{Y_t^2} + \frac{X_t (dY_t)^2}{Y_t^3} \\ &= \frac{f(X_t, t)dt + c(t)X_t dB_t}{Y_t} - \frac{X_t c(t)dB_t}{Y_t} - \frac{c(t)^2 X_t Y_t dt}{Y_t^2} + \frac{X_t c(t)^2 Y_t^2 dt}{Y_t^3} \\ &= \frac{f(X_t, t)}{Y_t} dt\end{aligned}$$

Say $z(t) = \frac{X_t}{Y_t}$ then

$$\frac{dz}{dt} = \frac{f(zY_t, t)}{Y_t}$$

this is an ODE to $z(t)$.

Example. Solve

$$dX_t = \frac{dt}{X_t} + \alpha X_t dB_t$$

First, find Y_t where

$$dY_t = \alpha Y_t dB_t$$

then

$$d(\log Y_t) = \frac{dY_t}{Y_t} - \frac{(dY_t)^2}{2Y_t^2} = \alpha dB_t - \frac{1}{2}\alpha^2 dt$$

then

$$Y_t = \exp \left(\alpha B_t - \frac{1}{2}\alpha^2 t \right)$$

in this case,

$$\begin{aligned}d\left(\frac{X_t}{Y_t}\right) &= \frac{1}{X_t Y_t} dt \\ \implies \frac{dz}{dt} &= \frac{1}{z Y_t^2} \\ \implies \int_{z_0}^z \tilde{z} d\tilde{z} &= \int_0^t \frac{1}{Y_s^2} ds\end{aligned}$$

$$\implies \frac{1}{2}(z^2 - z_0^2) = \int_0^t \exp(-2\alpha B_s + \alpha^2 s) ds$$

thus,

$$\begin{aligned} z &= (z_0^2 + 2 \int_0^t \exp(-2\alpha B_s + \alpha^2 s) ds)^{1/2} \\ \implies X_t &= \exp\left(\alpha B_t - \frac{1}{2}\alpha^2 t\right) (X_0^2 + 2 \int_0^t \exp(-2\alpha B_s + \alpha^2 s) ds)^{1/2} \end{aligned}$$

Example. (Lamperti transformation) Idea: Simplify SDEs by converting the major term to additive noise. Consider

$$dX_t = a(X_t)dt + b(X_t)dB_t$$

Find $Z_t = h(X_t)$ make the noise additive in the Z_t expression.

Use Itô calculus, then

$$\begin{aligned} dZ_t &= h'(X_t)dX_t + \frac{1}{2}h''(X_t)(dX_t)^2 \\ &= h'(X_t)(a(X_t)dt + b(X_t)dB_t) + \frac{1}{2}h''(X_t)b(X_t)^2dt \\ &= (a(X_t)h'(X_t) + \frac{1}{2}h''(X_t)b(X_t)^2)dt + h'(X_t)b(X_t)dB_t \end{aligned}$$

Need $h'(X_t)b(X_t) = 1$ then

$$h(x) = \int^x \frac{ds}{b(s)}$$

and

$$h'(x) = -\frac{b'}{b^2}$$

Substitute $h(x)$ then we have

$$dZ_t = \left(\frac{a(X_t)}{b(X_t)} - \frac{1}{2}b'(X_t)\right)dt + dB_t$$

Use the inverse function $X_t = h^{-1}(Z_t)$ then

$$dZ_t = \left(\frac{a(h^{-1}(Z_t))}{b(h^{-1}(Z_t))} - \frac{1}{2}b'(h^{-1}(Z_t))\right)dt + dB_t$$

which is an additive function well defined if $h^{-1}(Z_t)$ is well defined.

Example. (Cox-Ingersoll-Ross equation) (Models interest rates in some situations)

$$dR_t = (\alpha - \beta R_t)dt + \sigma\sqrt{R_t}dB_t$$

where $\alpha, \beta, \sigma > 0$.

$Z_t = h(R_t)$ where

$$\begin{aligned} h(r) &= \int^r \frac{ds}{\sigma\sqrt{s}} = \frac{2\sqrt{r}}{\sigma} \\ \implies r &= \left(\frac{\sigma Z_t}{2}\right)^2 = h^{-1}(Z_t) \end{aligned}$$

$a(r) = \alpha - \beta r$ then $a(h^{-1}(Z_t)) = \alpha - \frac{\beta\sigma^2 Z_t^2}{4}$. $b(r) = \sigma\sqrt{r}$ then $b(h^{-1}(Z_t)) = \frac{\sigma^2 Z_t}{2}$. $b'(r) = \frac{\sigma}{2\sqrt{r}}$ then $b'(h^{-1}(Z_t)) = \frac{1}{Z_t}$. In this case

$$\begin{aligned} dZ_t &= \left(\frac{2(\alpha - \frac{\beta\sigma^2 Z_t^2}{4})}{\sigma^2 Z_t} - \frac{1}{2Z_t}\right)dt + dB_t \\ &= \left(-\frac{\beta Z_t}{2} + \left(\frac{2\alpha}{\sigma^2} - \frac{1}{2}\right)\frac{1}{Z_t}\right)dt + dB_t \end{aligned}$$

Definition 1.14. (SDEs in n-dimensions)

$$d\underline{X}_t = \underline{a}(\underline{X}_t, t)dt + \underline{\beta}(\underline{X}_t, t)d\underline{B}_t$$

where $\underline{X}_t \in \mathbb{R}^n$, $\underline{a} \in \mathbb{R}^n$, $\underline{B}_t \in \mathbb{R}^n$ and $\underline{\beta} \in \mathbb{R}^n \mathbb{R}^n$.

Example. Dust particle in a turbulent flow

$$d\underline{X}_t = \underline{u}(\underline{X}_t, t)dt + \sqrt{2k}d\underline{B}_t$$

where $\underline{X}_t \in \mathbb{R}^2$. When $k = 0$ this is particle path equation and $\sqrt{2k}$ is diffusivity.

Itô's formula: Consider $f(\underline{X}_t, t)$ then

$$\begin{aligned} df &= f_t dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i} dX_t^{(i)} + \frac{1}{2}(\partial_{tt} f(dt)^2 + \sum_{i=1}^n \partial_{tx_i} dt dX_t^{(i)} + \sum_{i=1}^n \sum_{j=1}^n \partial_{x_i} \partial_{x_j} dX_t^{(i)} dX_t^{(j)}) \\ df &= f_t dt + \sum_{i=1}^n (a_i dt + \beta_{im} dB_t^{(m)}) \partial_{x_i} f + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \partial_{x_i x_j} f \beta_{im} \beta_{jn} \delta_{nm} dt \end{aligned}$$

Generator of SDE is

$$df = (\partial_t f + \underline{a} \cdot \nabla f + \frac{1}{2}(\underline{\beta\beta^T} : \nabla^2)f)dt + \nabla \cdot \underline{\beta} \cdot d\underline{B}_t$$

$$\mathcal{L}g = \underline{a} \cdot \nabla g + \frac{1}{2}(\underline{\beta\beta^T} : \nabla^2)g$$

2 SDEs to PDEs and basic

2.1 Transition density

Consider

$$dX_t = a(X_t, t)dt + b(X_t, t)dB_t, \quad X_s = y$$

and

$$B_t = \frac{1}{\sqrt{2\pi t^2}} \exp\left(-\frac{x^2}{2t}\right)$$

is the solution to the heat equation $\partial_t p = \frac{1}{2}\partial_{xx}p$.

Definition 2.1. The transition density $p_T(x, t|y, s)$ is the pdf $p(x, t)$ of X_t given that $X_s = y$.

Consider two types of problems:

- (1) How does $p(x, t)$ evolve when (y, s) are treated as fixed constants.
- (2) How does

$$u(y, s) = \mathbb{E}(g(X_t)|X_s = y)$$

behave

$$u(y, s) = \int_{\mathbb{R}} g(x)p_T(x, t|y, s) dx$$

Let $X_t = y + B_t - B_s$ be a Brownian process and

$$g(x) = \begin{cases} 1 & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

Interpretation $u(y, s)$ is $\mathbb{P}(a < X_t < b|X_s = y)$.

Examples. Properties of p_T :

- (a) (Markov property) Integrate the equation from s to t

$$X_t = X_s + \int_s^t a(X_u, u) du + \int_s^t b(X_u, u) dB_u$$

so $X_t|X_s$ is independent of X_u for $u < s$ and

$$p(x, t|y_1, s_1; \dots; y_n, s_n) = p_T(x, t|y, s)$$

where $s_1 < s_2 < \dots < s$ is the pdf version of Markov process.

- (b) (Chapman-Kolmogorov Property) Let $p(x, t, z, q|y, s)$ to be the joint pdf of $X_t = x$ and $X_q = z$. Then

$$\begin{aligned} p_T(x, t|y, s) &= \int_{\mathbb{R}} p(x, t, z, q|y, s) dz \\ &= \int_{\mathbb{R}} p(x, t|y, s; z, q)p_T(z, q|y, s) dz \\ &= \int_{\mathbb{R}} p_T(x, t|z, q)p_T(z, q|y, s) dz \end{aligned}$$

2.2 Fokker-Planck equations

How does $p(x, t) = p_T(x, t|y, s)$ with y, s fixed evolve in time?

$$dX_t = a(X_t, t)dt + b(X_t, t)dB_t, \quad X_0 = y$$

Consider $\mathbb{E}(f(X_t)) = \int_{\mathbb{R}} f(x)p(x, t) dx$. Idea: think about $\frac{d}{dt}$ of $\mathbb{E}(f(X_t))$ in more than one way:

(i)

$$\frac{d}{dt}\mathbb{E}(f(X_t)) = \int_{\mathbb{R}} f(x)p_t(x, t) dx$$

(ii)

$$\begin{aligned} df &= f'(X_t)dX_t + \frac{f''(X_t)}{2}(dX_t)^2 \\ &= f'(X_t)(a(X_t, t)dt + b(X_t, t)dB_t) + \frac{1}{2}f''(X_t)b(X_t, t)^2dt \end{aligned}$$

Integrate from 0 to t we have

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f'(X_s)a(X_s, s) ds + \frac{1}{2}f''(X_s)b(X_s, s)^2 ds + \int_0^t f'(X_s)b(X_s, s) dB_s \\ \implies \mathbb{E}(f(X_t)) &= \mathbb{E}(f(X_0)) + \mathbb{E}\left(\int_0^t f'(X_s)a(X_s, s) ds + \frac{1}{2}f''(X_s)b(X_s, s)^2 ds\right) \end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{d}{dt}\mathbb{E}(f(X_t)) = \mathbb{E}(f'(X_t)a(X_t, t) + \frac{1}{2}f''(X_t)b(X_t, t)^2) \\
&\frac{d}{dt}\mathbb{E}(f(X_t)) = \int_{\mathbb{R}} p(x, t)(f'(x)a(x, t) + \frac{1}{2}f''(x)b(x, t)^2) dx \\
&\Rightarrow \int_{\mathbb{R}} p(x, t)(f'(x)a(x, t) + \frac{1}{2}f''(x)b(x, t)^2) dx = \int_{\mathbb{R}} f(x)p_t(x, t) dx
\end{aligned}$$

Use the fact that for $p(x, t)$ to be a pdf $p_x \rightarrow 0$ as $|x| \rightarrow \infty$ then

$$\begin{aligned}
&\int_{\mathbb{R}} p(x, t)(f'(x)a(x, t) + \frac{1}{2}f''(x)b(x, t)^2) dx = \int_{\mathbb{R}} f(x)((-a(x, t)p(x, t))_x + (\frac{1}{2}b(x, t)^2p(x, t))_{xx}) \\
&\Rightarrow \int_{\mathbb{R}} f(x)(p_t + (ap)_x - \frac{1}{2}(b^2p)_{xx}) dx = 0
\end{aligned}$$

by Fundamental Lemma of variation

$$p_t + (ap)_x - \frac{1}{2}(b^2p)_{xx} = 0$$

which is known as the Fokker-Planck equations.

Example. The Brownian motion

$$dX_t = dB_t$$

satisfies $p_t = \frac{1}{2}p_{xx}$ has heat kernel solution

$$p(x, t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right)$$

Example. (Advection-diffusion)

$$dX_t = vdt + \sqrt{2\kappa}dB_t$$

then $p_t + vp_x = \kappa p_{xx}$.

Example. (O-U process)

$$dX_t = -\frac{X_t}{\tau}dt + \left(\frac{2\sigma^2}{\tau}\right)^{1/2}dB_t$$

then

$$p_t - \frac{(xp)_x}{\tau} = \frac{\sigma^2}{\tau}p_{xx}$$

this has a steady solution $p_s(x)$ then

$$(xp_s)' = \sigma^2 p_s''$$

and let $p_s \rightarrow 0$ as $|X| \rightarrow \infty$ then

$$\begin{aligned}
&C + xp_s = \sigma^2 p_s' \\
&\Rightarrow D - \frac{x^2}{2\sigma^2} = \log p_s \\
&\Rightarrow p_s = D \exp\left(-\frac{x^2}{2\sigma^2}\right) \\
&\Rightarrow p_s(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)
\end{aligned}$$

therefore

$$\Rightarrow \lim_{t \rightarrow \infty} X_t \sim N(0, \sigma^2)$$

Definition 2.2. A steady solution $p_s(x)$ of the FPE is called the invariant density of the SDE when $\lim_{t \rightarrow \infty} p(x, t) = p_s(x)$ for all initial conditions.

Example. $\mathcal{L}g = ag_x + \frac{1}{2}b^2g_{xx}$ if $p_t = \mathcal{L}^\dagger p$ then

$$\mathcal{L}^\dagger = -(ap)_x + \frac{1}{2}(b^2p)_{xx}$$

where $\mathcal{L}^\dagger$ is the adjoint of $\mathcal{L}$ with respect to the usual inner product $(f, g) = \int_{-\infty}^{\infty} f(x)g(x) dx$ and $\mathcal{L}^\dagger$ is defined so that

$$(\mathcal{L}f, g) = (f, \mathcal{L}^\dagger g)$$

for all $f, g \in L^2(\mathbb{R})$ under the decay boundary conditions.

(FPE initial conditions)

$$dX_t = vdt + \sqrt{2\kappa}dB_t \quad X_0 = y$$

what is the initial condition for $p(x, t)$.

$$p(x, 0) = \delta(x - y)$$

(FPE) boundary conditions. The FPE takes natural boundary conditions in applications:

1. (1) (Absorbing) In the SDE, the calculation is stopped when X_t hits X_+ . e.g. dust particle hits the wall, stock price hits payout value.
- (2) (Reflecting) In the SDE, the process is reflected at x_+ .

In the FPE, at absorbing $p(x_+, t) = 0$ and for the reflecting, notice that the FPE can be written

$$p_t + j_x = 0$$

where $j = -ap + \frac{1}{2}(b^2 p)_x$ and $j(x, t)$ is the probability current.

$$\frac{d}{dt} \left(\int_{x_-}^{x_+} p \, dx \right) = -j(x_+, t) + j(x_-, t)$$

since $\int_{x_-}^{x_+} p \, dx = 1$ so

$$j(x_-, t) = j(x_+, t) = 0$$

at reflecting boundary conditions. In absorbing case

$$p = \int_{x_-}^{x_+} p(x, t) \, dx$$

Here $1 - p$ is the probability that X_t has been absorbed.
(FPE n-dimensions)

$$d\underline{X}_t = \underline{a}(\underline{X}_t, t)dt + \underline{\beta}(\underline{X}_t, t) \cdot d\underline{B}_t$$

has FPE

$$p_t + \nabla \cdot (ap) = \frac{1}{2} \nabla \nabla : (\underline{\beta} \underline{\beta}^T p)$$

Example. For

$$d\underline{X}_t = d\underline{B}_t$$

then

$$\frac{\partial^2}{\partial x_i \partial x_j} (\delta_{ij} p) = \Delta p$$

If X_0 is a random variable then $p(x, 0) = p_0(x)$ is pdf of X_0 .

Example. FPE solutions methods

Difficult for general a, b so concentrate on constant coefficient case.

(1) (Heat kernel) $p_t + vp_x = \kappa p_{xx}$. Let $P(X, T) \equiv p(x, t)$ and let $X = x - vt$ then

$$\begin{aligned} P_T &= \kappa P_{xx} \\ \Rightarrow P(X, T) &= \frac{1}{\sqrt{4\pi\kappa T}} \exp\left(-\frac{(X-y)^2}{4\kappa T}\right) \\ \Rightarrow p(x, t) &= \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-vt-y)^2}{4\kappa t}\right) \end{aligned}$$

(2) (The method of image)

Example. Consider

$dX_t = dB_t$ $X_0 = y > 0$
and X_t is absorbed at $x = 0$ then $p_t = \frac{1}{2}p_{xx}$, $p(x, 0) = \delta(x - y)$ and $p(0, t) = 0$

$$\Rightarrow p(x, t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x-y)^2}{2t}\right) - \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x+y)^2}{2t}\right)$$

For reflecting, $p_x(0, t) = 0$ here use the positive image

$$\Rightarrow p(x, t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x-y)^2}{2t}\right) + \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x+y)^2}{2t}\right)$$

(3) (Fourier transform)

$$p(x, t) = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} p_0(x - q) \exp\left(-\frac{q^2}{2t}\right) dq$$

Convolution integral between initial data and heat kernel.

(4) (Separation of variables)

$$p_t = \kappa p_{xx}$$

with boundary conditions. Need to use $p(x, t) = X(x)T(t)$ use boundary condition in X to find series solutions.

2.3 The backwards Kolmogorov equation

Given

$$dX_t = a(X_t, t)dt + b(X_t, t)dB_t, \quad X_s = y$$

Consider $u(y, s) = \mathbb{E}(g(X_t)|X_s = y)$, we want to derive an equation for $u(y, s)$. Note that

$$u(y, s) = \int_{-\infty}^{\infty} g(x) p_T(x, t|y, s) \, dx$$

Lemma 2.3. (Markov property)

$$\mathbb{E}(u(X_\tau, \tau)|X_s = y) = u(y, s)$$

for all $s \leq \tau \leq t$. (Expected value of $g(X_t)$ does not change as the process runs).

Proof.

$$\begin{aligned}
\mathbb{E}(u(X_\tau, \tau) | X_s = y) &= \int_{-\infty}^{\infty} u(z, \tau) p_T(z, \tau | y, s) dz \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x) p_T(x, t | z, \tau) dx \right) p_T(z, \tau | y, s) dz \\
&= \int_{-\infty}^{\infty} g(x) \left(\int_{-\infty}^{\infty} p_T(x, t | z, \tau) p_T(z, \tau | y, s) dz \right) dx \\
&= \int_{-\infty}^{\infty} g(x) p_T(x, t | y, s) dx = u(y, s)
\end{aligned}$$

□

At $t = s$ the original equation becomes

$$dX_s = a(y, s)dt + b(y, s)dB_s$$

Take the Taylor expansion on $u(y + dX_s, s + dt)$ we have

$$\begin{aligned}
u(y + dX_s, s + dt) &= u(y, s) + u_s(y, s)dt + u_y(y, s)dX_s + \frac{1}{2}(u_{yy}(y, s)(dX_s)^2) \\
&= u(y, s) + (u_s(y, s) + u_y(y, s)a(y, s) + \frac{1}{2}u_{yy}(y, s)b(y, s)^2)dt + (u_y(y, s)b(y, s))dB_s
\end{aligned}$$

Take the expectation we have

$$\begin{aligned}
u(y, s) &= u(y, s) + (u_s + au_y + \frac{1}{2}u^2u_{yy})dt \\
&\implies u_s + au_y + \frac{1}{2}b^2u_{yy} = 0
\end{aligned}$$

which is known as the backwards Kolmogorov equation. This is a parabolic equation but s is not "time like". But it becomes diffusion-like when you use $\tilde{t} = t_0 = s$. BKE takes a final condition $u(y, t) = g(y)$.

Example. Consider

$$dX_t = vdt + \sqrt{2\kappa}dB_t, \quad X_0 = y$$

Say $g(x) = \chi[a, b]$

$$\begin{aligned}
\mathbb{P}(y) &= \mathbb{P}(X_T \in [a, b] | X_0 = y) \\
&= \mathbb{E}(g(X_t) | X_0 = y) \\
u(y, s) &= \mathbb{E}(g(X_t) | X_s = y)
\end{aligned}$$

then

$$u_s + vu_y + \kappa u_{yy} = 0, \quad u(y, \tau) = g(y)$$

Introduce $\tilde{t} = \tau - s$ yields

$$\begin{aligned}
\tilde{u}(y, \tilde{t}) &= \tilde{u}(y, \tau - s) = u(y, s) \\
&\implies \tilde{u}_{\tilde{t}} - v\tilde{u}_y = \kappa\tilde{u}_{yy}
\end{aligned}$$

Use $\tilde{y} = y + v\tilde{t}$ yields

$$\begin{aligned}
w(\tilde{y}, \tilde{t}) &= w(y + v\tilde{t}, \tilde{t}) = \tilde{u}(y, \tilde{t}) \\
&\implies w_{\tilde{t}} = \kappa w_{\tilde{y}\tilde{y}}, \quad w(\tilde{y}, 0) = g(\tilde{y})
\end{aligned}$$

Use Fourier transform solution

$$\begin{aligned}
w(\tilde{y}, \tilde{t}) &= \frac{1}{\sqrt{4\pi\kappa\tilde{t}}} \int_{-\infty}^{\infty} g(z) \exp\left(-\frac{(z - \tilde{y})^2}{4\kappa\tilde{t}}\right) dz \\
&= \frac{1}{\sqrt{4\pi\kappa\tilde{t}}} \int_a^b \exp\left(-\frac{(z - \tilde{y})^2}{4\kappa\tilde{t}}\right) dz \\
&= \frac{1}{2} \left(\operatorname{erf}\left(\frac{\tilde{y} - a}{\sqrt{4\kappa\tilde{t}}}\right) - \operatorname{erf}\left(\frac{\tilde{y} - b}{\sqrt{4\kappa\tilde{t}}}\right) \right) \\
&\implies u(y, s) = \frac{1}{2} \left(\operatorname{erf}\left(\frac{y + v(\tau - s) - a}{\sqrt{4\kappa(\tau - s)}}\right) - \operatorname{erf}\left(\frac{y + v(\tau - s) - b}{\sqrt{4\kappa(\tau - s)}}\right) \right)
\end{aligned}$$

Multiple dimension version of the backwards Kolmogorov equation is

$$u_s + \underline{a} \cdot \nabla u + \frac{1}{2}(\underline{\beta} \cdot \underline{\beta}^T : \nabla \nabla)u = 0$$

2.4 Feynman-Kac Formula

Consider

$$u(y, s) = \mathbb{E}(g(X_t) \exp\left(-\int_s^t q(X_r, r) dr\right) | X_s = y)$$

Aim: Generalise BKE by finding equation for u . Notice that $q = 0$ recovers the BKE situation. Problem in this case is that $u(u, s)$ does not satisfy Markov lemma.

Idea: Consider joint SDE

$$\begin{aligned} dX_t &= a(X_t, t)dt + b(X_t, t)dB_t, & X_s &= y \\ dZ_t &= -Z_t q(X_t, t)dt, & Z_s &= z \end{aligned}$$

Try

$$\begin{aligned} d(\log Z_t) &= \frac{dZ_t}{Z_t} - \frac{1}{2} \frac{(dZ_t)^2}{Z_t^2} \\ &= -q(X_t, t)dt \\ \implies \log Z_t - \log Z_s &= \int_s^t -q(X_r, r) dr \\ \implies Z_t &= Z_s \exp\left(-\int_s^t q(X_r, r) dr\right) \end{aligned}$$

Define

$$\begin{aligned} v(y, z, s) &= \mathbb{E}(g(X_t)Z_t | X_s = y, Z_s = z) \\ &= \mathbb{E}(g(X_t)Z_s \exp\left(-\int_s^t q(X_r, r) dr\right) | X_s = y, Z_s = z) \\ &= zu(y, s) \end{aligned}$$

Notice $v(y, z, s)$ satisfies all conditions for BKE in 2D then

$$\underline{a} = \begin{pmatrix} a(y, t) \\ -zq(y, t) \end{pmatrix}, \quad \underline{\beta} = \begin{pmatrix} b(y, t) & 0 \\ 0 & 0 \end{pmatrix}$$

then

$$\begin{aligned} v_s + av_y + (-zq)v_z + \frac{1}{2}b^2v_{yy} &= 0 \\ \implies zu_s + azu_y + (-zq)u &= \frac{1}{2}b^2zu_{yy} = 0 \\ \implies u_s + au_y + \frac{1}{2}b^2u_{yy} - qu &= 0 \end{aligned}$$

which is known as the Feynman-Kac equation. Substitute $s = t$ yields

$$u(y, t) = \mathbb{E}(g(X_t) | X_t = y) = g(y)$$

and FK operator is $u_s + (\mathcal{L} - q)u = 0$.

Example. (Pricing "European options") Consider

$$dX_t = rX_t dt + \sigma X_t dB_t, \quad X_s = y$$

then

$$u(y, s) = \mathbb{E}(g(X_t)e^{-r_0(\tau-s)} | X_s = y)$$

where

$$g(x) = \begin{cases} x - k & x \geq k \\ 0 & \text{otherwise} \end{cases}$$

and k is the strike price and r_0 is market interest rate.

This is FK problem with $q = r_0$, $a(y) = ry$ and $b(y) = \sigma(y)$. Then

$$u_s + ryu_y + \frac{1}{2}\sigma^2 y^2 u_{yy} - r_0 u = 0$$

where $g(y, \tau) = g(y)$.

Example. (Chemistry on a particle) Consider

$$dX_t = vdt + \sqrt{2\kappa}dB_t, \quad X_0 = y$$

Particle random-walk in 1D. Contains chemical C_t , undergoes a decay at rate $q(X < t)$

$$dC_t = q(X_t, t)C_t dt, \quad C_0 = c_0$$

What is the expected amount at chemical in $a < x < b$ at time τ ?

We need

$$\begin{aligned} Q &= \mathbb{E}(C_\tau g(X_\tau) | X_0 = y, C_0 = c_0) \\ C_t &= c_0 \exp\left(-\int_0^t q(X_r, r) dr\right) \\ Q &= c_0 \mathbb{E}(g(X_t) \exp\left(-\int_0^t q(X_r, r) dr\right) | X_0 = y) \\ u(y, s) &= \mathbb{E}(g(X_\tau) \exp\left(-\int_s^\tau q(X_r, r) dr\right) | X_s = y) \\ \implies Q &= c_0 u(y, 0) \end{aligned}$$

so $u(y, s)$ obeys F-K

$$u_s + vu_y + \kappa u_{yy} - q(y, s)u = 0 \quad u(y, \tau) = g(y)$$

Example. (Radioactive Particle) Decay rate $q(x, t)$. Let P_t be $\mathbb{P}(\text{Particle still exists})$ then

$$dP_t = -q(X_t, t)P_t dt$$

$\mathbb{P}(\text{Finding undecayed particle in } [a, b])$

is the same as equation above with $c = 1$.

2.5 First Passage Time Problems

Definition 2.4. $T(\underline{y})$ is the first time, the process $\underline{X}_t$ reaches the boundary of D . $T(\underline{y}) = \inf\{t : X_t \in \partial D\}$.

Of interest,

$$\bar{T}(\underline{y}) = \mathbb{E}(T | \underline{X}_0 = \underline{y})$$

which is the mean first passage time. nth moment of T :

$$M_n(\underline{y}) = \mathbb{E}(T^n | \underline{X}_0 = \underline{y})$$

and $\rho(\underline{y}, t)$ is the PDF of T with $\int_0^\infty \rho(\underline{y}, t) dt = 1$. Time for stock to go bankrupt, dust particle to escape, protein to fold. We focus on autonomous SDEs

$$dX_t = a(X_t)dt + b(X_t)dB_t$$

Preliminary results

(i) Time-lag property

$$\mathbb{E}((f(T)|X_s = y) = \mathbb{E}(f(T + s)|X_0 = y))$$

To see this mathematically: let $\rho(y, t)$ be the PDF of T with $X_0 = y$. Now consider $X_s = y$ then

$$dX_t = a(X_t)dt + b(X_t)dB_t, \quad X_s = y$$

Define $\tilde{t} = t - s$ then $\tilde{X}_{\tilde{t}} = X_{\tilde{t}+s} = X_t$ and $\tilde{B}_{\tilde{t}} = B_t - B_s$. Thus,

$$d\tilde{X}_{\tilde{t}} = a(\tilde{X}_{\tilde{t}})d\tilde{t} + b(\tilde{X}_{\tilde{t}})d\tilde{B}_{\tilde{t}}, \quad \tilde{X}_0 = y$$

which implies PDF of T is $\rho(y, t - s)$ and

$$\mathbb{E}(f(T)|X_s = y) = \int_s^\infty f(t)\rho(y, t - s) dt = \int_0^\infty f(\tilde{t} + s)\rho(y, \tilde{t}) d\tilde{t} = \mathbb{E}(f(T + s)|X_0 = y)$$

(ii) (Law of iterated expectation) X, Y are random variables in some probability space then

$$\mathbb{E}(X) = \mathbb{E}_0(\mathbb{E}(X|Y))$$

Consider $\bar{T}(y) = \mathbb{E}(T|X_0 = y)$ then $dX_0 = a(y)dt + b(y)dB_0$ which is the first step of the process then

$$\begin{aligned} \mathbb{E}(T|X_0 = y) &= \mathbb{E}_0(\mathbb{E}(T|X_0 = y, X_{dt} = y + dX_0)) = \mathbb{E}_0(\mathbb{E}(T|X_{dt} = y + dX_0)) \\ &= \mathbb{E}_0(\mathbb{E}(T + dt|X_0 = y + dX_0)) = dt + \mathbb{E}_0(\mathbb{E}(T|X_0 = y + dX_0)) = dt + \mathbb{E}(\bar{T}(y + dX_0)) \\ &\implies \bar{T} = dt + \mathbb{E}(\bar{T} + \bar{T}'(y)dX_0 + \frac{1}{2}\bar{T}''(y)(dX_0)^2) \\ &= dt + \mathbb{E}(\bar{T}(y) + \bar{T}'(y)(a(y)dt + b(y)dB_0) + \frac{1}{2}\bar{T}''(y)b(y)^2 dt) \\ \bar{T}(y) &= dt + \bar{T}(y) + \bar{T}'(y)a(y)dt + \frac{1}{2}\bar{T}''(y)b(y)^2 dt \end{aligned}$$

So our mean first passage time equation is

$$a\bar{T}' + \frac{1}{2}b^2\bar{T}'' = -1$$

the operator form is

$$\mathcal{L}\bar{T} = -1$$

with $\bar{T}(x_\pm) = 0$.

Example. Consider

$$dX_t = vdt + \sqrt{2\kappa}dB_t \quad X_0 = y \in (0, 1)$$

What is the expected time for X_t to hit 0 or 1?

Our mean first passage time equation is

$$\begin{aligned} a\bar{T}' + \kappa\bar{T}'' &= -1, \quad \bar{T}(0) = \bar{T}(1) = 0 \\ \kappa(\bar{T}'e^{vy/\kappa})' &= -e^{vy/\kappa} \\ \implies \kappa\bar{T}'e^{vy/\kappa} &= -\frac{\kappa}{v}e^{vy/\kappa} + C \\ \bar{T}' &= -\frac{1}{v} + De^{-vy/\kappa} \\ \implies \bar{T} &= -\frac{y}{v} + Ee^{-vy/\kappa} + F \end{aligned}$$

since $\bar{T}(0) = \bar{T}(1) = 0$ we know that

$$\begin{aligned} E + F &= 0 \\ -\frac{1}{v} + F(1 - e^{-v/\kappa}) &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow E &= \frac{1}{v(e^{-v/\kappa} - 1)}, \quad F = \frac{1}{v(1 - e^{-v/\kappa})} \\ \Rightarrow \bar{T}(y) &= -\frac{y}{v} + \frac{1}{v} \left(\frac{1 - e^{-yv/\kappa}}{1 - e^{-v/\kappa}} \right) \end{aligned}$$

If $|\frac{v}{\kappa}| \gg 1$ then the expected time is

$$\bar{T} \approx \begin{cases} \frac{1-y}{v} & v > 0 \\ -\frac{y}{v} & v < 0 \end{cases}$$

Example. Consider

$$dX_t = -\frac{X_t}{\tau} dt + \left(\frac{2\sigma^2}{\tau}\right)^{1/2} dB_t, \quad X_0 = y$$

Find the expected time $\bar{T}$ for the process to first hit $|X_t| = q > |y|$.

The MFPT equation is

$$-\frac{y}{\tau} \bar{T}' + \frac{\sigma^2}{\tau} \bar{T}'' = -1, \quad \bar{T}(\pm q) = 0$$

Using integration factor yields

$$\sigma^2 (\bar{T}' e^{-y^2/2\sigma^2})' = -\tau e^{-y^2/2\sigma^2}$$

Recall $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds$. Then

$$\bar{T}' e^{-y^2/2\sigma^2} = -\frac{\tau}{\sigma^2} \int_0^y e^{-t^2/2\sigma^2} dt + D$$

then

$$\begin{aligned} \bar{T}' e^{-y^2/2\sigma^2} &= -\frac{\tau}{\sigma} \sqrt{\frac{\pi}{2}} \text{erf}\left(\frac{y}{\sqrt{2}\sigma}\right) \\ \bar{T}' &= -\frac{\tau}{\sigma} \sqrt{\frac{\pi}{2}} \text{erf}\left(\frac{y}{\sqrt{2}\sigma}\right) e^{y^2/2\sigma^2} + D e^{y^2/2\sigma^2} \end{aligned}$$

since $\bar{T}$ is even so $D = 0$ then

$$\bar{T}(y) = \tau \sqrt{\pi} \left(F\left(\frac{q}{\sqrt{2}\sigma}\right) - F\left(\frac{y}{\sqrt{2}\sigma}\right) \right)$$

where $F(z) = \int_0^z \text{erf}(s) e^{s^2} ds$.

MFPT in operator form is

$$\mathcal{L}\bar{T} = -1$$

and in n-dimensions,

$$\underline{a} \cdot \nabla_y \bar{T} + \frac{1}{2} (\underline{\Sigma} : \nabla_y \nabla_y) \bar{T} = -1$$

where $\underline{\Sigma} = \beta \beta^T$.

(Reflection boundary conditions) Suppose the process is reflected at $X_t = x_-$ and absorbed at $X_t = x_+$. $\bar{T}$ is the first time process hits $X_T = x_+$. MFPT here is

$$a\bar{T}' + \frac{1}{2} b^2 \bar{T}'' = -1$$

Recall

$$\begin{aligned} \bar{T}(y) &= dt + \mathbb{E}(\bar{T}(y + dX_0)) \\ dX_0 &= a(y)dt + b(y)dB_0 \end{aligned}$$

then

$$\begin{aligned} \bar{T}(y) &= dt + (dt + \bar{T}' dX_0 + \frac{1}{2} \bar{T}'' (dX_0)^2) \\ 0 &= dt + \mathbb{E}(\bar{T}'(y)(a(y)dt + b(y)dB_0) + \frac{1}{2} \bar{T}''(y)b^2(y)dt) \end{aligned}$$

When $y = x_-$, dB_0 is forced to be $+ve$. Then

$$\mathbb{E}(dB_0|_{y=x_-}) = (dt^2)$$

Moments and distribution of T

$$dX_t = a(X_t)dt + b(X_t)dB_t, \quad X_0 = y$$

T is first time X_t reaches x_- or x_+ . Define $\rho(y, t)$ to be pdf of T and $M_n(y) = \int_0^\infty t^n \rho(y, t) dt$ is the nth moment. There are two methods:

1. (Method 1) Consider

$$v(y, s) = \mathbb{E}(e^{-sT} | X_0 = y)$$

this is related to Moment generating function: $\mathbb{E}(e^{sT} | X_0 = y) = v(y, -s)$. Notice that

$$v(y, s) = \int_0^\infty e^{-st} \rho(y, t) dt$$

i.e. $v(y, s)$ is the Laplace transform of ρ . If we know $v(y, s)$ then

$$\rho(y, t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} v(y, s) e^{st} ds$$

which is known as the Mellin inverse formula. Can also get $M_n(y)$ directly from $v(y, s)$ by

$$\frac{\partial^n v}{\partial s^n} = \mathbb{E}\left(\frac{\partial^n}{\partial s^n} (e^{-sT}) | X_0\right) = \mathbb{E}((-1)^n T^n e^{-sT} | X_0 = y)$$

Set $s = 0$ we have

$$\begin{aligned}
\frac{\partial^n v}{\partial s^n}(y, 0) &= (-1)^n \mathbb{E}(T^n | X_0 = y) = (-1)^n M_n(y) \\
v(y, s) &= \mathbb{E}(e^{-sT} | X_0 = y) \\
&= \mathbb{E}_0(\mathbb{E}(e^{-sT} | X_0 = y, X_{dt} = y + dX_0)) \\
&= \mathbb{E}_0(\mathbb{E}(e^{-s(T+dt)} | X_0 = y + dX_0)) \\
&= e^{-sdt} \mathbb{E}_0(\mathbb{E}(e^{-sT} | X_0 = y + dX_0)) \\
&= e^{-sdt} \mathbb{E}_0(v(y + dX_0, s)) \\
&= (1 - sdt) \mathbb{E}_0(v(y, s) + v_y(y, s)dX_0 + \frac{1}{2}v_{yy}(y, s)(dX_0)^2) \\
&= (1 - sdt)(v(y, s) + a(y)v_y dt + b^2(t)v_{yy}dt) \\
&\implies \frac{1}{2}b^2v_{yy} + av_y - sv = 0
\end{aligned}$$

T is first time X_t reaches x_- or x_+ so $v(x_{\pm}, s) = 1$ is the boundary condition. The operator form is in fact $\mathcal{L}v - sv = 0$ which gives the formula in higher dimensions.

Example. Let

$$dX_t = \sqrt{2}dB_t, \quad X_0 = y$$

and T is the first time $X_t = \pm q$.

The Laplace transform equation in this case is

$$v_{yy} - sv = 0, \quad v(\pm q, s) = 1$$

so we need $s > 0$ for v to exist. Then

$$v(y, s) = C \cosh(\sqrt{s}y) + D \sinh(\sqrt{s}y)$$

By the boundary condition $v(\pm q, s) = 1$ we have $D = 0$ and $C = \frac{1}{\cosh(\sqrt{s}q)}$ thus

$$v(y, s) = \frac{\cosh(\sqrt{s}y)}{\cosh(\sqrt{s}q)}$$

The inverse Laplace transform is hard but at least we can determine

$$\bar{T} = M_1 = -\frac{\partial v}{\partial s}(y, 0) = \frac{1}{2}(q^2 - y^2)$$

Similarly,

$$\text{Var}(T) = M_2 - M_1^2 = \frac{1}{6}(q^4 - y^4)$$

2. (Method 2) The moment equation

$$aM'_n(y) + \frac{1}{2}b^2(y)M''_n(y) = -nM_{n-1}(y)$$

use to get $M_1 = T, M_2, \dots$ etc.

3. (Method 3) Idea: Solve

$$p_t + (ap)_x = \frac{1}{2}(b^2p)_{xx}$$

with boundary condition $p(x_{\pm}, t) = 0, p(x, 0) = \delta(x - y)$. Consider

$$\mathbb{P}(\text{has been absorbed}) = P(y, t) = 1 - \int_{x_-}^{x_+} p(x, t) dx = \mathbb{P}(T \leq t)$$

so $P(y, t)$ is actually the cumulative density function(cdf) of T . Thus,

$$\rho(y, t) = \frac{\partial}{\partial t}P(y, t) = - \int_{x_-}^{x_+} p_t(x, t) dx = \int_{x_-}^{x_+} j_x(x, t) dx = j(x_+, t) - j(x_-, t)$$

where $p_t + j_x = 0$ and $j = ap - \frac{1}{2}(b^2p)_x$ is the probability current.

2.6 Hitting problem

Idea: Often we are not interested in T but in $g(X_t)$, the value at some function X_t at time T when it hits ∂D_1 .

Example.

$$g(x) = \begin{cases} 1 & x \in \partial\Omega_1 \\ 0 & x \in \partial\Omega_2 \end{cases}$$

Then $\psi(y) = \mathbb{E}(g(X_T) | X_0 = y)$ is $\mathbb{P}(\text{process first hit } \partial D \text{ on } \partial D_1)$.

$$dX_t = a(X_t)dt + b(X_t)dB_t$$

then

$$\mathbb{E}(g(X_T) | X_\tau = y) = \mathbb{E}(g(X_T) | X_0 = y) = \psi(y)$$

so probability exiting through ∂D_1 is not affected by short time.

Use first step method again we have

$$\begin{aligned}
\psi(y) &= \mathbb{E}(g(X_T)|X_0 = y) \\
&= \mathbb{E}_0(\mathbb{E}(g(X_T)|X_0 = y, X_{dt} = y + dX_0)) \\
&= \mathbb{E}_0(\psi(y + dX_0)) \\
&= \psi(y) + dt(a(y)\psi' + \frac{1}{2}b(y)^2\psi'')
\end{aligned}$$

So we have our hitting equation

$$a(y)\psi' + \frac{1}{2}b(y)^2\psi'' = 0$$

with boundary conditions $\psi(x_+) = g(x_+) = 1$, $\psi(x_-) = g(x_-) = 0$.

Example. Consider

$$dX_t = vdt + \sqrt{2\kappa}dB_t$$

where $\psi(1) = 1$ and $\psi(0) = 0$.

The hitting equation is

$$\begin{aligned}
v\psi' + \kappa\psi'' &= 0 \\
\kappa(\psi'e^{yv/\kappa})' &= 0 \\
\psi'e^{yv/\kappa} &= C \\
\implies \psi &= De^{-yv/\kappa} + E \\
\implies \psi(y) &= \frac{1 - e^{-yv/\kappa}}{1 - e^{-v/\kappa}}
\end{aligned}$$

3 Approximation methods for SDEs

3.1 Adiabatic removal of fast variables(homogenisation)

Example. (Brownian particle) Let $0 < \varepsilon \ll 1$

$$\begin{aligned} m\ddot{x} &= -r\dot{x} + \text{'noise'} \implies \\ dX_t &= \frac{V_t}{\varepsilon} dt \\ dV_t &= -\frac{V_t}{\varepsilon^2} dt + \left(\frac{2}{\varepsilon^2}\right)^{1/2} dB_t \end{aligned}$$

Our aim is to average over the 'fast' process (O-U process for V_t , timescale $\varepsilon^2 \ll 1$ to get 'slow' equation for X_t valid on long timescales (here $O(1)$). Start by considering the FPE for $p(x, v, t)$ (pdf of (X_t, V_t))

$$p_t + \nabla \cdot (\underline{a}p) = \frac{1}{2} \nabla \nabla : (\underline{\Sigma}p)$$

where

$$\begin{aligned} \nabla &\equiv \begin{pmatrix} \partial_x \\ \partial_v \end{pmatrix}, \quad \underline{a} = \begin{pmatrix} v/\varepsilon \\ -v/\varepsilon^2 \end{pmatrix}, \quad \underline{\beta} = \left(\frac{2}{\varepsilon^2}\right)^{1/2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ \implies \underline{\beta}\underline{\beta}^T &= \left(\frac{2}{\varepsilon^2}\right) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Thus the FPE is

$$p_t + \left(\frac{v}{\varepsilon}p\right)_x + \left(-\frac{v}{\varepsilon^2}p\right)_v = \left(\frac{1}{\varepsilon^2}\right)p_{vv}$$

which is known as Kramer's equation. In order to solve this equation, exploit the fact that $\varepsilon \ll 1$. Seek a solution $p(x, v, t; \varepsilon) = p_0(x, v, t) + \varepsilon p_1(x, v, t) + O(\varepsilon^2)$. Insert in Kramer's equation we have the leading order

$$-(vp_0)_v = (p_0)_{vv}$$

can integrate:

$$-vp_0 = (p_0)_v + C(x, t)$$

p_0 is part of pdf so we can use boundary condition that $p_0, p_0 \rightarrow 0$ as $|v| \rightarrow \infty$ so $C(x, t) = 0$. Then

$$\begin{aligned} -vp_0 = (p_0)_v &\implies \log p_0 = -\frac{v^2}{2} + \log P(x, t) \\ p_0 &= P(x, t)e^{-v^2/2} \end{aligned}$$

When we look at the next order at $O(\varepsilon)$ the equation is

$$(vp_0)_x = (vp_1)_v + (p_1)_{vv}$$

this equation has a solvability condition because

$$\int_{\mathbb{R}} (vp_1)_v + (p_1)_{vv} dv = 0$$

which gives

$$\int_{\mathbb{R}} (vp_0)_x dv = \int_{\mathbb{R}} P_x(x, t)ve^{-v^2/2} dv = 0$$

Still need to solve

$$(vp_1)_v + (p_1)_{vv} = P_x(x, t)ve^{-v^2/2}$$

Try $p_1 = F(x, t)ve^{-v^2/2}$ then

$$\begin{aligned} (p_1)_v + vp_1 &= Fe^{-v^2/2} \implies (p_1)_{vv} + (vp_1)_v = -Fve^{-v^2/2} \\ \implies -F &= P_x(x, t) \end{aligned}$$

therefore,

$$p_1 = -P_x(x, t)ve^{-v^2/2} + P_1(x, t)e^{-v^2/2}$$

At $O(1)$,

$$(p_2)_{vv} + (vp_2)_v = (vp_1)_x + (p_0)_t$$

No need to solve, just use the solvability condition yields

$$\begin{aligned} \int_{\mathbb{R}} (vp_1)_x + (p_0)_t dv &= 0 \\ \implies \int_{\mathbb{R}} (-vP_xve^{-v^2/2})_x + (vP_1e^{-v^2/2})_x + (Pe^{-v^2/2})_t dv &= 0 \end{aligned}$$

therefore,

$$\begin{aligned} -P_{xx} \int_{\mathbb{R}} v^2e^{-v^2/2} dv + (P_1)_x \int_{\mathbb{R}} ve^{-v^2/2} dv + P_t \int_{\mathbb{R}} e^{-v^2/2} dv &= 0 \\ \implies P_t - P_{xx} &= 0 \end{aligned}$$

corresponds to the SDE

$$dX_t = \sqrt{2}dW_t$$

known as the slow dynamics equation. Conclusion, viewed on long times the Brownian particle obeys a Brownian process.

Example. (Dimensional Langevin) Consider the equation

$$m\ddot{x} = -r\dot{x} + \gamma\eta(t)$$

where $\eta(t) = \frac{dB_t}{dt}$ informally. If $\dot{x} = \bar{V}_{\bar{t}}$ and $x = \bar{X}_{\bar{t}}$ then

$$d\bar{V}_{\bar{t}} = -\frac{r}{m}\bar{V}_{\bar{t}}d\bar{t} + \frac{\gamma}{m}d\bar{B}_{\bar{t}}$$

$$d\bar{X}_{\bar{t}} = \bar{V}_{\bar{t}}d\bar{t}$$

substitute $\tau = m/r$, $\sigma^2 = \frac{\gamma^2}{2mr}$, $V_t = \frac{\bar{V}_{\bar{t}}}{\sigma}$, $t = \frac{\bar{t}}{t_0}$ where $\varepsilon^2 = \frac{\tau}{t_0} \ll 1$ then

$$\sigma dV_t = -\frac{r\sigma}{m}V_t t_0 dt + \frac{\gamma}{m}d\bar{B}_{t t_0}$$

$$\implies dV_t = -\frac{V_t}{\varepsilon^2}dt + \frac{\sqrt{2}dB_t}{\varepsilon}$$

$$dX_t = \frac{V_t}{\varepsilon}dt$$

then

$$dX_t = \sqrt{2}dW_t$$

$$d\bar{X}_t = \sqrt{2\kappa}d\bar{W}_{\bar{t}}$$

where $\kappa = \frac{m\sigma^2}{r}$.

Example. (Haken's system) Consider

$$dX_t = (-X_t + \frac{a}{\varepsilon}X_t Y_t)dt + \sqrt{2\kappa}dB_t$$

$$dY_t = (-\frac{Y_t}{\varepsilon^2} + \frac{bX_t^2}{\varepsilon})dt + \frac{\sqrt{2}}{\varepsilon}dB_t$$

the FPE is

$$p_t = \frac{1}{\varepsilon^2}\mathcal{L}_0^\dagger p + \frac{1}{\varepsilon}\mathcal{L}_1^\dagger p + \mathcal{L}_2^\dagger p$$

Insert asymptotic expansion:

$$p = p_0 + \varepsilon p_1 + \varepsilon^2 p_2 + \dots$$

At $O(\varepsilon^{-2})$ we have

$$\mathcal{L}_0^\dagger p_0 = 0$$

At $O(\varepsilon^{-1})$ we have

$$\mathcal{L}_0^\dagger p_1 = -\mathcal{L}_1^\dagger p_0$$

At $O(1)$ we have

$$\mathcal{L}_0^\dagger p_2 = -\mathcal{L}_1^\dagger p_1 - \mathcal{L}_2^\dagger p_0 + p_0 t$$

Here FPE is

$$\begin{aligned} p_t + \nabla \cdot (ap) &= \frac{1}{2}\nabla\nabla : (\Sigma p) \\ p_t &= ((x + \frac{a}{\varepsilon}xy)p)_x + ((-\frac{y}{\varepsilon^2} + \frac{bx^2}{\varepsilon})p)_y + \kappa p_{xx} + \frac{1}{\varepsilon^2}p_{yy} \\ \mathcal{L}_0^\dagger p &= (yp)_y + p_{yy} \\ \mathcal{L}_1^\dagger p &= (axy p)_x - (bx^2 p)_y \\ \mathcal{L}_2^\dagger p &= \kappa p_{xx} + (xp)_x \end{aligned}$$

at $O(\varepsilon^{-2})$ we have

$$\implies p_0 = P(x, t)e^{-y^2/2}$$

at $O(\varepsilon^{-1})$ we have

$$(yp_1)_y + (p_1)_{yy} = -(axy p_0)_x + (bx^2 p_0)_y$$

check the solvability condition yields

$$\int_{\mathbb{R}} (axy P(x, t)e^{-y^2/2})_x dy = (axP)_x \int_{\mathbb{R}} ye^{-y^2/2} dy = 0$$

try $p_1 = F(x, t)ye^{-y^2/2}$ then

$$\begin{aligned} yp_1 + (p_1)_y &= Fe^{-y^2/2} \\ (yp_1)_y + (p_1)_{yy} &= -Fye^{-y^2/2} \\ \implies F &= (axP)_x + bx^2P \end{aligned}$$

so

$$p_1 = ((axP)_x + bx^2P)ye^{-y^2/2} + P_1e^{-y^2/2}$$

At $O(1)$ we have

$$\begin{aligned} \mathcal{L}_0^\dagger p_2 &= -\mathcal{L}_1^\dagger p_1 - \mathcal{L}_2^\dagger p_0 + (p_0)_t \\ (yp_2)_y + (p_2)_{yy} &= (-axy p_1)_x + (bx^2 p_1)_y - (xp_0)_x - \kappa(p_0)_{xx} + (p_0)_t \end{aligned}$$

Use solvability condition,

$$0 = -(axF)_x \int_{\mathbb{R}} y^2 e^{-y^2/2} dy - (xP)_x \int_{\mathbb{R}} e^{-y^2/2} dy + (axP_1)_x \int_{\mathbb{R}} ye^{-y^2/2} dy - \kappa P_{xx} \int_{\mathbb{R}} e^{-y^2/2} dy + P_t \int_{\mathbb{R}} e^{-y^2/2} dy$$

so

$$\begin{aligned} P_t - \kappa P_{xx} - (xP)_x - (axF)_x &= 0 \\ P_t &= \kappa P_{xx} + (ax(axP)_x + abx^3P)_x \\ P_t + ((a^2 - 1)x - abx^3)P &= ((\kappa + a^2x^2)P)_{xx} \end{aligned}$$

which is the FPE of

$$dX_t = ((a^2 - 1)X_t - abX_t^3)dt + \sqrt{2}\sqrt{\kappa + a^2X_t^2}dB_t$$

Example. A general problems where $0 < \varepsilon \ll 1$,

$$\begin{aligned} dX_t &= \left(\frac{1}{\varepsilon}a(X_t, Y_t) + b(X_t, Y_t)\right)dt + c(X_t, Y_t)dB_{1t} \\ dY_t &= \frac{1}{\varepsilon^2}f(X_t, Y_t)dt + \frac{1}{\varepsilon}g(X_t, Y_t)dB_{2t} \end{aligned}$$

Aim: eliminate fast process Y_t and obtain the slow dynamics expression for X_t . The FPE is given by

$$p_t = -\left(\frac{1}{\varepsilon}a(x, y) + b(x, y)p\right)_x - \frac{1}{\varepsilon^2}(fp)_y + \frac{1}{2}(c^2p)_{xx} + \frac{1}{2\varepsilon^2}(g^2p)_{yy}$$

or

$$p_t = \frac{1}{\varepsilon^2}\mathcal{L}_0^\dagger p + \frac{1}{\varepsilon}\mathcal{L}_1^\dagger p + \mathcal{L}_2^\dagger p$$

Insert $p = p_0 + \varepsilon p_1 + O(\varepsilon^2)$ then

$$\begin{aligned} \mathcal{L}_0^\dagger p_0 &= 0 \\ \mathcal{L}_0^\dagger p_1 &= -\mathcal{L}_1^\dagger p_0 \\ \mathcal{L}_0^\dagger p_2 &= -\mathcal{L}_1^\dagger p_1 - \mathcal{L}_2^\dagger p_0 + p_t \end{aligned}$$

then

$$\mathcal{L}_0^\dagger p_0 \equiv -(fp_0)_y + \frac{1}{2}(g^2p_0)_{yy} = 0$$

which is a steady FPE of the dynamics given by

$$dY_t = f(x, Y_t)dt + g(x, Y_t)dB_t$$

x fixed here is a parameter. Need the fast dynamics to have an invariant density for all $x \in \mathbb{R}$, so that $\mathcal{L}_0^\dagger p_0 = 0$ has a normalised solution i.e.

$$\mathcal{L}_0^\dagger p_s(x, y) = 0 \quad \forall x$$

and

$$\int_{\mathbb{R}} p_s(x, y) dy = 1$$

then we solve the equation $(fp_s)_y = \frac{1}{2}(g^2p_s)_{yy}$ which gives

$$\begin{aligned} fp_s &= \frac{1}{2}(g^2p_s)_y \\ \implies p_s &= \frac{P_0}{g^2} \exp\left(\int^y \frac{2f}{g^2} dq\right) \end{aligned}$$

so

$$P_0^{-1} = \int_{\mathbb{R}} \frac{1}{g^2} \exp\left(\int^y \frac{2f}{g^2} dq\right) dy$$

Proceed by writing $p_0(x, y, t) = P(x, t)p_s(x, y)$. Equation $\mathcal{L}_0^\dagger p_1 = \mathcal{L}_1^\dagger p_0$ is based on centering condition on $a(x, y)$ where

$$\int_{\mathbb{R}} a(x, y)p_s(x, y) dy = 0$$

i.e. $\mathbb{E}_F(a(x, Y_t)) = 0$. The third equation is a bit nasty but quite similar.

3.2 Activation energy problems

Example. (Motion of particle in potential) Let $0 < \varepsilon \ll 1$

$$dX_t = -V'(X_t)dt + (2\varepsilon)^{1/2}dB_t$$

the FPE is given by

$$p_t - (V'(x)p)_x = \varepsilon p_{xx}$$

the steady FPE is $-(V'(x)p_s)' = \varepsilon p_s''$ then

$$\begin{aligned} V'(x)p_s &= \varepsilon p_s' \implies \log p_s = -\frac{V(x)}{\varepsilon} + \log P_0 \\ \implies p_s &= P_0 \exp\left(-\frac{V(x)}{\varepsilon}\right) \end{aligned}$$

where

$$P_0 = \left(\int_{\mathbb{R}} \exp\left(-\frac{V(x)}{\varepsilon}\right) dx\right)^{-1}$$

therefore, $V(x) \rightarrow \infty$ for $|x| \rightarrow \infty$ for p_s to exist. What is the typical behaviour? $\varepsilon \ll 1 \implies$ small oscillations about

$x = x_A$ or x_B . For smooth $V(x)$ we have

$$V'(x) = V'(x_A) + (x - x_A)V''(x_A) + \dots$$

gives

$$dX_t = -(X_t - x_A)V''(x_A)dt + (2\varepsilon)^{1/2}dB_t$$

make substitution $Q_t = X_t - x_A$ gives

$$dQ_t = -Q_t V''(x_A)dt + (2\varepsilon)^{1/2}dB_t = -\frac{Q_t}{\tau}dt + \left(\frac{2\sigma^2}{\tau}\right)^{1/2}dB_t$$

so the decorrelation time is $\tau = \frac{1}{V''(x_*)}$ and $\sigma = (\varepsilon V''(x_*))^{1/2} > 0$. The standard deviation is small which justifies our approximation.

Example. (The Kramers (first passage time) problem) How long is the transition time from $A \rightarrow B$? Introducing reflecting boundary condition at $x = x_-$ and absorbing boundary condition at $x = x_+$. The MFPT is given by

$$-V'(y)\bar{T}'(y) + \varepsilon\bar{T}''(y) = -1$$

where $\bar{T}'(x_-) = 0$ and $\bar{T}(x_+) = 0$. Then

$$\varepsilon(\bar{T}'e^{-V(y)/\varepsilon})' = -e^{-V(y)/\varepsilon}$$

integrating gives

$$\varepsilon\bar{T}'e^{-V(y)/\varepsilon} = -\int_y^x e^{-V(q)/\varepsilon} dq$$

Use $\bar{T}'(x_-) = 0$ gives

$$\begin{aligned}\bar{T}'e^{-V(y)/\varepsilon} &= -\frac{1}{\varepsilon} \int_{x_-}^y e^{-V(q)/\varepsilon} dq \\ \Rightarrow \bar{T}'(y) &= -\frac{1}{\varepsilon} e^{V(y)/\varepsilon} \int_{x_-}^y e^{-V(q)/\varepsilon} dq \\ \Rightarrow \bar{T}(y) &= -\frac{1}{\varepsilon} \int_{x_+}^y e^{V(z)/\varepsilon} \left(\int_{x_-}^z e^{-V(q)/\varepsilon} dz \right) dz = \frac{1}{\varepsilon} \int_y^{x_+} e^{V(z)/\varepsilon} \left(\int_{x_-}^z e^{-V(q)/\varepsilon} dz \right) dz\end{aligned}$$

Can we explicit $\varepsilon \ll 1$ to get a more interesting result so we need to approximate integrals.

Proposition 3.1. (Laplace's Method) Consider $I(s) = \int_{t_-}^{t_+} f(t)e^{sg(t)} dt$ where $f(t)$, $g(t)$ are smooth functions on $[t_-, t_+]$. We are interested in the behaviour as $s \rightarrow \infty$. $g(t)$ has either an isolated global maximum at $t = t_0 \in (t_-, t_+)$ or a maximum at endpoint $t = t_{\pm}$ with $|g'(t_{\pm})| \neq 0$. For the first case, integrand is largest near $t = t_0$ and decays rapidly when $s \rightarrow \infty$, then

$$\begin{aligned}I(s) &\approx \int_{t_-}^{t_+} (f(t_0) + (t - t_0)f'(t_0) + \dots) \exp\left(s(g(t_0) + (t - t_0)g'(t_0) + \frac{1}{2}(t - t_0)^2g''(t_0))\right) dt \\ &= e^{sg(t_0)} \int_{t_-}^{t_+} (f(t_0) + (t - t_0)f'(t_0) + \dots) \exp\left(\frac{1}{2}(t - t_0)^2sg''(t_0) + \dots\right) dt\end{aligned}$$

Let $q = (t - t_0)\sqrt{s}$ then

$$\begin{aligned}I(s) &\approx e^{sg(t_0)} \int_{(t_- - t_0)\sqrt{s}}^{(t_+ - t_0)\sqrt{s}} \left(f(t_0) + \frac{q}{\sqrt{s}}f'(t_0) + \dots\right) \exp\left(-\frac{1}{2}q^2|g''(t_0)| + O\left(\frac{1}{\sqrt{s}}\right)\right) \frac{dq}{\sqrt{s}} \\ &= f(t_0)e^{sg(t_0)} \left(\frac{2\pi}{s|g''(t_0)|}\right)^{1/2}\end{aligned}$$

as $s \rightarrow \infty$. Similarly, for case 2

$$I(s) \approx f(t_{\pm})e^{sg(t_{\pm})} \frac{1}{s|g'(t_{\pm})|}$$

We look back to

$$\bar{T}(y) = \frac{1}{\varepsilon} \int_y^{x_+} e^{V(z)/\varepsilon} J(z, \varepsilon) dz$$

where $J(z, \varepsilon) = \int_{x_-}^z e^{-V(t)/\varepsilon} dt$ and $z \in [y, x_+]$. Use Laplace's method with $f(t) = 1$, $g(t) = -V(t)$ and $s = \frac{1}{\varepsilon}$ then

$$\begin{aligned}J(z, \varepsilon) &\sim e^{-V(x_A)/\varepsilon} \left(\frac{2\pi\varepsilon}{|V''(x_A)|}\right)^{1/2} \\ \Rightarrow \bar{T}(y) &= \frac{1}{\varepsilon} \left(\frac{2\pi\varepsilon}{|V''(x_A)|}\right)^{1/2} e^{-V(x_A)/\varepsilon} \int_y^{x_+} e^{V(t)/\varepsilon} dt \\ &= \frac{1}{\varepsilon} \left(\frac{2\pi\varepsilon}{|V''(x_A)|}\right)^{1/2} e^{-V(x_A)/\varepsilon} \cdot e^{V(x_M)/\varepsilon} \left(\frac{2\pi\varepsilon}{|V''(x_M)|}\right)^{1/2} \\ \Rightarrow \bar{T}(y) &= \frac{2\pi}{(|V''(x_A)V''(x_M)|)^{1/2}} \exp\left(\frac{V(x_M) - V(x_A)}{\varepsilon}\right)\end{aligned}$$

which is known as the Krammer's time. $\bar{T}$ depends almost entirely on the potential difference between x_M and x_A . And

similarly the time for $B \rightarrow A$ is given by

$$\bar{T}(y) = \frac{2\pi}{(|V''(x_B)V''(x_M)|)^{1/2}} \exp\left(\frac{V(x_M) - V(x_B)}{\varepsilon}\right)$$

The time taken to reach x_c is given by

$$\bar{T}(y) = \frac{(2\pi\varepsilon)^{1/2}}{(|V''(x_c)V''(x_A)|)^{1/2}} \exp\left(\frac{V(x_c) - V(x_A)}{\varepsilon}\right)$$

3.3 Hamiltonian systems with noise

Definition 3.2. A dynamical system in $2N$ dimensions $\underline{x} \in \mathbb{R}^{2N}$

$$\frac{d\underline{x}}{dt} = \underline{F}(\underline{x}, t)$$

is called Hamiltonian if $(x_1, \dots, x_N, x_{N+1}, \dots, x_{2N})$ can be transformed into variables $(q_1, \dots, q_N, p_1, \dots, p_N)$ so that the equation becomes

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i} \quad i = 1, \dots, N$$

which is called the Hamilton's equations and $H(q_1, \dots, q_N, p_1, \dots, p_N)$ is the Hamilton. $(q_1, \dots, q_N)$ is called spatial coordinates and $(p_1, \dots, p_N)$ are called conjugate momenta. The $\mathbb{R}^{2N}$ space spanned by $(q_1, \dots, q_N, p_1, \dots, p_N)$ is called phase-space.

Example. Consider the SHM

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0$$

Let $q = y$ and $p = \frac{dy}{dt}$ then

$$\frac{dq}{dt} = p, \quad \frac{dp}{dt} = -\omega^2 q$$

this is Hamiltonian with $H(q, p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$.

Example. Consider motion in a potential force

$$\frac{d^2 \underline{y}}{dt^2} = -\nabla V(\underline{y}), \quad \underline{y} \in \mathbb{R}^3$$

Let

$$\underline{q} = (q_1, \dots, q_N) = (y_1, \dots, y_N), \quad \underline{p} = (p_1, \dots, p_N) = \left(\frac{dy_1}{dt}, \dots, \frac{dy_N}{dt}\right)$$

then

$$\frac{d\underline{q}}{dt} = \underline{p}, \quad \frac{d\underline{p}}{dt} = -\nabla_{\underline{q}} V(\underline{q})$$

therefore,

$$H = \frac{1}{2}(p_1^2 + \dots + p_N^2) + V(q_1, \dots, q_N)$$

so H is often energy.

In general, $H = H(q_1, \dots, q_N, p_1, \dots, p_N, t)$ then

$$\frac{dH}{dt} = \frac{\partial H}{\partial q_i} \cdot \frac{dq_i}{dt} + \frac{\partial H}{\partial p_i} \cdot \frac{dp_i}{dt} + \frac{\partial H}{\partial t}$$

by Hamilton's equations

$$\frac{dH}{dt} = \frac{\partial H}{\partial q_i} \left(\frac{\partial H}{\partial p_i}\right) + \frac{\partial H}{\partial p_i} \left(-\frac{\partial H}{\partial q_i}\right) + \frac{\partial H}{\partial t}$$

so H is conserved as long as $\frac{\partial H}{\partial t} = 0$. We are going to consider $N = 1$ for simplicity.

Definition 3.3. (The cycle average) We define the cycle average as follows

$$\bar{f}(T) = \frac{1}{T} \int_0^T f(q(t), p(t)) dt$$

Here $f(q, p)$ is any function defined on the phase-space. T is the period of an oscillation.

Recall

$$\oint_{C(h)} F(q, p) ds = \int_0^T F(q(t), p(t)) \frac{ds}{dt} dt$$

as $0 < t < T \implies C(h)$. Then if we choose $F = f/(ds/dt)$ we have

$$\bar{f}(h) = \frac{1}{T} \oint_{C(h)} \frac{f(p, q)}{ds/dt} ds$$

But we know that

$$\frac{ds}{dt} = \sqrt{\left(\frac{dp}{dt}\right)^2 + \left(\frac{dq}{dt}\right)^2} = ((-H_q)^2 + (H_p)^2)^{1/2} = |\nabla_{\mathbf{p}} H|$$

thus,

$$\bar{f}(H) = \frac{1}{T} \oint_{C(h)} \frac{f(q, p)}{|\nabla_{\mathbf{p}} H|} ds$$

Note that

$$\begin{aligned} T(h) &= \int_0^T dt = \oint_{C(h)} \frac{ds}{|\nabla_{\mathbf{p}} H|} \\ \Rightarrow \bar{f}(h) &= \oint_{C(h)} \frac{f(q, p)}{|\nabla_{\mathbf{p}} H|} ds / \oint_{C(h)} \frac{ds}{|\nabla_{\mathbf{p}} H|} \end{aligned}$$

Perturbed system is denoted by $\begin{pmatrix} Q_t \\ P_t \end{pmatrix}$. Our aim is to derive equation for evolution of $H_t = H(Q_t, P_t)$.

Example. Consider the Perturbed system

$$d\underline{P}_t = (-\nabla^\perp H(\underline{P}_t) + \varepsilon \underline{F}(\underline{P}_t))dt + \varepsilon^{1/2} \underline{\beta} \cdot d\underline{B}_t$$

where $\nabla_{\mathbf{p}}^\perp = \begin{pmatrix} -\partial p \\ \partial q \end{pmatrix}$. Apply Itô to H_t we have

$$dH_t = (\underline{a} \cdot \nabla_{\mathbf{p}} H + \frac{1}{2} \varepsilon (\underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}}) H) dt + \varepsilon^{1/2} \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot d\underline{B}_t$$

where $\underline{a} = -\nabla_{\mathbf{p}}^\perp H + \varepsilon \underline{F}$ note that $\underline{a} \cdot \nabla_{\mathbf{p}} H = \varepsilon \underline{F} \cdot \nabla_{\mathbf{p}} H$ Thus,

$$dH_t = (\varepsilon \underline{F} \cdot \nabla_{\mathbf{p}} H + \frac{1}{2} \varepsilon (\underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}}) H) dt + \varepsilon^{1/2} \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot d\underline{B}_t$$

Our idea is to time-average this equation over time τ where $1 \ll \tau \ll \varepsilon^{-1}$. Average are many periods but H_t is almost constant. Integrating our equation gives

$$H_{t+\tau} - H_t = \varepsilon \int_t^{t+\tau} \underline{F} \cdot \nabla_{\mathbf{p}} H + \frac{1}{2} (\underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}}) H dr + \varepsilon^{1/2} \int_t^{t+\tau} \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot d\underline{B}_r$$

Note that all $\mathbf{p}$ are evaluated at $\underline{p}_r$ then if we replace $\underline{p}_r$ in the integrands with the deterministic trajectory $\underline{p}(r)$, which begins at $\underline{p}(t) = \mathbf{p}$ and obeys the $\varepsilon = 0$ equations. Then we can use cycle averages to replace time-average time-averages

$$\varepsilon \tau \cdot \frac{1}{\tau} \int_t^{t+\tau} \underline{F} \cdot \nabla_{\mathbf{p}} H + \frac{1}{2} (\underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}}) H dr = \varepsilon \tau \bar{a}(H_t)$$

with $\bar{a}(h) = \frac{1}{T(h)} \oint_{C(h)} \frac{\underline{F} \cdot \nabla_{\mathbf{p}} H + \frac{1}{2} \underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}} H}{|\nabla_{\mathbf{p}} H|} ds$. For noise integral

$$\varepsilon^{1/2} \int_t^{t+\tau} \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot d\underline{B}_r$$

By law of addition

$$\begin{aligned} \int_0^t \underline{f}(s) \cdot d\underline{B}_s &\sim N(0, \int_0^t |\underline{f}(s)|^2 ds) \\ \Rightarrow \varepsilon^{1/2} \int_t^{t+\tau} \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot d\underline{B}_r &\sim N(0, \varepsilon \int_t^{t+\tau} |\nabla_{\mathbf{p}} H \cdot \underline{\beta}|^2 dr) \\ |\underline{f}|^2 &= |\nabla_{\mathbf{p}} H \cdot \underline{\beta}|^2 = \nabla_{\mathbf{p}} H \cdot \underline{\beta} \cdot \underline{\beta}^T \cdot \nabla_{\mathbf{p}} H = \nabla_{\mathbf{p}} H \cdot \underline{\Sigma} \cdot \nabla_{\mathbf{p}} H \end{aligned}$$

Thus, if

$$\bar{b}(H) = \frac{1}{T(h)} \oint_{C(h)} \frac{\nabla_{\mathbf{p}} H \cdot \underline{\Sigma} \cdot \nabla_{\mathbf{p}} H}{|\nabla_{\mathbf{p}} H|} ds$$

then since

$$\begin{aligned} N(0, \int_t^{t+\tau} |\underline{f}|^2 dr) &\sim (\frac{1}{\tau} \int_t^{t+\tau} |\underline{f}|^2 dr)^{1/2} (W_{t+\tau} - W_t) \\ H_{t+\tau} - H_t &= \varepsilon \tau \bar{a}(H_t) + \varepsilon^{1/2} \bar{b}(H_t)^{1/2} (W_{t+\tau} - W_t) \end{aligned}$$

Replace $\tau \rightarrow dt$ we have

$$dH_t = \varepsilon \bar{a}(H_t) dt + \varepsilon^{1/2} \bar{b}(H_t)^{1/2} dW_t$$

which is known as the Hamiltonian evolution equation.

Example. (Perturbed sum) Let $0 < \varepsilon \ll 1$

$$\ddot{y} + \omega^2 y = -\mu y + \text{'noise'} \Rightarrow$$

$$dQ_t = P_t dt$$

$$dP_t = (-\omega^2 Q_t - \varepsilon P_t) dt + (\varepsilon)^{1/2} \sigma dB_t$$

Recall that $H(q, p) = \frac{1}{2}(\omega^2 q^2 + p^2)$ we know that

$$dH_t = \varepsilon \bar{a}(H_t) dt + \varepsilon^{1/2} \bar{b}(H_t)^{1/2} dW_t$$

need to find $\bar{a}$ and $\bar{b}$. We first find $T(h)$,

$$T(h) = \oint_{C(h)} \frac{ds}{|\nabla_{\mathbf{p}} H|}$$

then if

$$q(\theta) = \frac{\sqrt{2h}}{\omega} \cos \theta, \quad p(\theta) = \sqrt{2h} \sin \theta, \quad \theta \in (-\pi, \pi)$$

and note that $|\nabla_{\mathbf{p}} H| = \sqrt{\omega^4 q^2 + p^2}$ so $|\nabla_{\mathbf{p}} H| = \sqrt{2h}(\omega^2 \cos^2 \theta + \sin^2 \theta)^{1/2}$ and

$$\begin{aligned} ds &= \left(\left(\frac{dp}{d\theta} \right)^2 + \left(\frac{dq}{d\theta} \right)^2 \right)^{1/2} d\theta = \sqrt{2h}(\cos^2 \theta + (-\frac{1}{\omega} \sin \theta)^2) d\theta \\ \implies \frac{ds}{|\nabla_{\mathbf{p}} H|} &= \frac{\sqrt{2h}(\cos^2 \theta + (-\frac{1}{\omega} \sin \theta)^2) d\theta}{\sqrt{2h}(\omega^2 \cos^2 \theta + \sin^2 \theta)^{1/2}} = \frac{d\theta}{\omega} \end{aligned}$$

Therefore,

$$T(h) = \int_{-\pi}^{\pi} \frac{d\theta}{\omega} = \frac{2\pi}{\omega}$$

which is the period of simple harmonic motion. Then note that

$$\begin{aligned} \underline{F} &= \begin{pmatrix} 0 \\ -p \end{pmatrix}, \quad \underline{\beta} = \begin{pmatrix} 0 & 0 \\ 0 & \sigma \end{pmatrix}, \quad \underline{\Sigma} = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 \end{pmatrix} \\ \implies \underline{F} \cdot \nabla_{\mathbf{p}} H &= \begin{pmatrix} 0 \\ -p \end{pmatrix} \cdot \begin{pmatrix} \omega^2 q \\ p \end{pmatrix} = -p^2 \\ \frac{1}{2}(\underline{\Sigma} : \nabla_{\mathbf{p}\mathbf{p}} H) &= \frac{1}{2}\sigma^2 H_{pp} = \frac{1}{2}\sigma^2 \\ \implies \nabla_{\mathbf{p}} H \cdot \underline{\Sigma} \cdot \nabla_{\mathbf{p}} H &= \sigma^2 p^2 \end{aligned}$$

Therefore,

$$\begin{aligned} \bar{a}(h) &= \frac{1}{T} \oint_{C(h)} \frac{F \cdot \nabla_{\mathbf{p}} H + \frac{1}{2}(\underline{\Sigma} : \nabla_{\mathbf{p}} \nabla_{\mathbf{p}})}{|\nabla_{\mathbf{p}} H|} ds = \frac{\omega}{2} \int_{-\pi}^{\pi} (-2h \sin^2 \theta + \frac{1}{2}\sigma^2) \frac{d\theta}{\omega} \\ \implies \bar{a}(h) &= -h + \frac{1}{2}\sigma^2 \end{aligned}$$

Similarly,

$$\begin{aligned} \bar{b}(H) &= \frac{1}{T} \oint_{C(h)} (\nabla_{\mathbf{p}} H \cdot \underline{\Sigma} \cdot \nabla_{\mathbf{p}} H) \cdot \frac{ds}{|\nabla_{\mathbf{p}} H|} \\ &= \frac{\omega}{2\pi} \int_{-\pi}^{\pi} \sigma^2 \cdot 2h \sin^2 \theta \frac{d\theta}{\omega} = \sigma^2 h \end{aligned}$$

Thus, our noisy SHM is

$$dH_t = \varepsilon(-H_t + \frac{1}{2}\sigma^2)dt + \varepsilon^{1/2}\sigma\sqrt{H_t}dW_t$$

4 Linear Response Theory

Concept: Use the statistics of a stationary stochastic process to predict the response of the system to a force. We will consider the autonomous SDE

$$d\mathbf{X}_t^0 = \underline{a}(\mathbf{X}_t^0)dt + \underline{\beta}(\mathbf{X}_t^0) \cdot d\mathbf{B}_t \quad (*)$$

Assume $\mathbf{X}_t^0$ is stationary i.e. it has invariant density $p_s(\underline{x})$. We have access to unlimited observations of $\mathbf{X}_t^0$. So we can estimate $p_s(\underline{x})$ and other statistics.

Example. O-U process is an example for such process where

$$dX_t^0 = -\frac{X_t^0}{\tau} + \left(\frac{2\sigma^2}{\tau}\right)^{1/2} d\mathbf{B}_t$$

and

$$p_s = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$$

Example. Consider the perturbed system

$$d\mathbf{X}_t^\varepsilon = (\underline{a}(\mathbf{X}_t^\varepsilon) + \varepsilon \underline{F}(\mathbf{X}_t^\varepsilon, t))dt + \underline{\beta}(\mathbf{X}_t^\varepsilon) \cdot d\mathbf{B}_t$$

assume $f(\underline{x}, t) = f(\underline{x})\rho(t)$ is separable.

Our aim is to express the response

$$R_{f,g}(t) = \mathbb{E}(g(\mathbf{X}_t^\varepsilon) - g(\mathbf{X}_t^0))$$

in terms of the statistics of $\mathbf{X}_t^0$. Here $g(x)$ is called the observable.

Definition 4.1. A correlation function $C_{y,z}(t, s)$ between two stochastic processes Y_t and Z_t is

$$C_{y,z}(t, s) = \mathbb{E}(Y_t Z_s) \quad 0 \leq s < t$$

Suppose the joint process (Y_t, Z_t) is stationary (e.g. functions of the solutions of an autonomous SDE). Then recall the time-lag property $p_T(x, t|y, s) = p(x, t-s|y, 0)$ we have

$$C_{y,z}(\tau) = \mathbb{E}(Y_\tau Z_0), \quad \tau = t - s$$

If $Y_t = a(\mathbf{X}_t^0)$ and $Z_t = b(\mathbf{X}_t^0)$ then

$$\begin{aligned} C_{a,b}(t, s) &= \mathbb{E}(a(\mathbf{X}_t^0)b(\mathbf{X}_s^0)) \\ &= \int_D \int_D a(\underline{x})b(\underline{y})p(\underline{x}, \underline{y}, t, s) d\underline{x}d\underline{y} \end{aligned}$$

Denote $p_T^0(\underline{x}, t|\underline{y}, s)$ be the pdf of $\mathbf{X}_t^0|\mathbf{X}_s^0$ then

$$C_{a,b}(t, s) = \int_D \int_D a(\underline{x})b(\underline{y})p_T^0(\underline{x}, t|\underline{y}, s)p_s(\underline{y}) d\underline{x}d\underline{y}$$

and

$$C_{a,b}(\tau) = \int_D \int_D a(\underline{x})b(\underline{y})p_T^0(\underline{x}, \tau|\underline{y}, 0)p_s(\underline{y}) d\underline{x}d\underline{y}$$

To proceed consider $R_{f,g}(t) = \mathbb{E}(g(\mathbf{X}_t^\varepsilon) - g(\mathbf{X}_t^0))$ then by definition

$$\int g(\underline{x})(p(\underline{x}, t) - p_s(\underline{x})) d\underline{x}$$

Use iterated expectation

$$\begin{aligned} \mathbb{E}(g(\mathbf{X}_t^\varepsilon)) &= \mathbb{E}_0(\mathbb{E}(g(\mathbf{X}_t^\varepsilon)|\mathbf{X}_0^\varepsilon)) = \mathbb{E}_0\left(\int_D p_T(\underline{x}, t|\mathbf{X}_0^\varepsilon, 0)g(\underline{x}) d\underline{x}\right) \\ &= \int_D \int_D g(\underline{x})p_T(\underline{x}, t|\underline{y}, 0)p_s(\underline{y}) d\underline{x}d\underline{y} \\ &\implies p(\underline{x}, t) = \int_D p_T(\underline{x}, t|\underline{y}, 0)p_s(\underline{y}) d\underline{y} \end{aligned}$$

Here $p(\underline{x}, t)$ is the pdf of $\mathbf{X}_t^\varepsilon$ so it satisfies the FPE

$$\begin{aligned} p_t &= \mathcal{L}_0^\dagger p + \varepsilon \rho(t) \mathcal{L}_1^\dagger p \\ &= -\nabla \cdot (\underline{a}p) + \frac{1}{2} \nabla \nabla : (\underline{\Sigma}p) - \varepsilon \rho(t) \nabla \cdot (\underline{f}_s p) \end{aligned}$$

with the initial condition $p(\underline{x}, 0) = p_s(\underline{x})$. We shall seek a solution of the form $p = p_0 + \varepsilon p_1 + \varepsilon^2 p_2 + O(\varepsilon^3)$ then at $O(1)$ we have

$$p_{0t} = \mathcal{L}_0^\dagger p_0, \quad p_0(\underline{x}, 0) = p_s(\underline{x})$$

solution is $p_0 = p_s(\underline{x})$. At $O(\varepsilon)$ we have

$$p_{1t} - \mathcal{L}_0^\dagger p_1 = \rho(t) \mathcal{L}_1^\dagger p_s, \quad p_1(\underline{x}, 0) = 0$$

Use green function method

$$q_t + \mathcal{L}_0^\dagger q = \delta(t)(\mathcal{L}_1^\dagger p_s), \quad q(\underline{x}, 0^-) = 0$$

Lemma 4.2. If $q(\underline{x}, t)$ is known as above then

$$p_1(\underline{x}, t) = \int_{-\infty}^{\infty} \rho(s) q(\underline{x}, t-s) ds$$

Proof. We evaluate the equation as follows

$$\begin{aligned} (\partial_t - \mathcal{L}_0^\dagger) \left(\int_{-\infty}^{\infty} \rho(s) q(\underline{x}, t-s) ds \right) &= \int_{-\infty}^{\infty} \rho(s) (\partial_t - \mathcal{L}_0^\dagger) q(\underline{x}, t-s) ds \\ &= \int_{-\infty}^{\infty} \rho(s) (\partial_t - \mathcal{L}_0^\dagger) q(\underline{x}, t) ds = \int_{-\infty}^{\infty} \rho(s) [\delta(\tau) \mathcal{L}_1^\dagger p_s] ds = \rho(t) \mathcal{L}_1^\dagger p_s \end{aligned}$$

□

Now we find $q(\underline{x}, t)$ we integrate from 0^- to 0^+ to get

$$q(\underline{x}, 0^+) = \mathcal{L}_1^\dagger p_s, \quad q_t - \mathcal{L}_0^\dagger q = 0$$

this solution can be therefore written in terms of the transition density p_T^0

$$q(\underline{x}, t) = \int_D p_T^0(\underline{x}, t | \underline{y}, 0) (\mathcal{L}_1^\dagger p_s)(\underline{y}) d\underline{y}$$

Recall that $p_T^0(\underline{x}, t | \dots)$ is a solution of the FPE also satisfies initial conditions since $p_T^0(\underline{x}, 0 | \underline{y}, 0) = \delta(\underline{x}, \underline{y})$. Then

$$\begin{aligned} R_{f,g}(t) &= \int_D g(\underline{x}) (p(\underline{x}, t) - p_s(\underline{x})) d\underline{x} \\ &= \int_D g(\underline{x}) (p_0 + \varepsilon p_1 + O(\varepsilon^2) - p_s) ds \\ &= \varepsilon \int_D g(\underline{x}) \left(\int_{-\infty}^{\infty} \rho(s) q(\underline{x}, t-s) ds \right) d\underline{x} + O(\varepsilon^2) \\ &= \varepsilon \int_{-\infty}^{\infty} \rho(s) \chi(t-s) ds + O(\varepsilon^2) \end{aligned}$$

where

$$\chi(t) = \int_D q(\underline{x}, t) g(\underline{x}) d\underline{x} = \int_D \int_D g(\underline{x}) p_T^0(\underline{x}, 0 | \underline{y}, 0) (\mathcal{L}_1^\dagger p_s)(\underline{y}) d\underline{x} d\underline{y}$$

or

$$\chi(t) = C_{g,h}(t)$$

with

$$h(\underline{x}) = \frac{\mathcal{L}_1^\dagger p_s}{p_s} = -p_s^{-1} \nabla \cdot (f_s p_s)$$

Example. (2023 Q5) Consider with $0 < \varepsilon \ll 1$

$$dX_t^\varepsilon = \left(-\frac{X_t^\varepsilon}{\tau} + \varepsilon f(t) \right) dt + \left(\frac{2\sigma^2}{\tau} \right)^{1/2} dB_t$$

- (a) What is the invariant density of $\varepsilon = 0$.
- (b) Find X_t^ε in terms of X_0^ε .
- (c) Let $R(t) = \mathbb{E}((X_t^\varepsilon)^3 - (X_t^0)^3)$. Show that

$$R(t) = \varepsilon \int_{-\infty}^{\infty} \chi(s) f(t-s) ds + O(\varepsilon^2)$$

and find $\chi(t)$.

- (d) Use the FDT to express $\chi(t)$ as a correlation function involving X_t^0 .

- (a) This is given by

$$p_s(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-x^2/2\sigma^2}$$

- (b) Note that

$$d(X_t^\varepsilon e^{t/\tau}) e^{t/\tau} dX_t^\varepsilon + \frac{X_t^\varepsilon}{\tau} e^{t/\tau} dt = e^{t/\tau} \varepsilon f dt + \left(\frac{2\sigma^2}{\tau} \right)^{1/2} dB_t$$

Integrating gives

$$\begin{aligned} X_t^\varepsilon e^{t/\tau} - X_0^\varepsilon &= \varepsilon \int_0^t f(s) e^{s/\tau} ds + \left(\frac{2\sigma^2}{\tau} \right)^{1/2} \int_0^t e^{s/\tau} dB_s \\ \implies X_t^\varepsilon &= X_t^0 e^{-t/\tau} + \varepsilon \int_0^t f(s) e^{-(t-s)/\tau} ds + \left(\frac{2\sigma^2}{\tau} \right)^{1/2} \int_0^t e^{-(t-s)/\tau} dB_s \end{aligned}$$

- (c) $X_t^\varepsilon = X_t^0 + \varepsilon \int_0^t f(s) e^{-(t-s)/\tau} ds$ then

$$(X_t^\varepsilon)^3 = (X_t^0 + \varepsilon \int_0^t f(s) e^{-(t-s)/\tau} ds)^3 = (X_t^0)^3 + 3(X_t^0)^2 \varepsilon \int_0^t f(s) e^{-(t-s)/\tau} ds + O(\varepsilon^2)$$

so

$$\begin{aligned}\mathbb{E}((X_t^\varepsilon)^3 - (X_t^0)^3) &= 3\mathbb{E}((X_t^0)^2)\varepsilon \int_0^t f(s)e^{-(t-s)/\tau} ds + O(\varepsilon^2) \\ \implies R(t) &= \varepsilon(3\sigma^2 \int_0^t f(s)e^{-(t-s)/\tau} ds) + O(\varepsilon^2)\end{aligned}$$

so

$$\chi(t) = 3\sigma^2 e^{-t/\tau} H(t)$$

(d) Since $\chi(t) = C_{g,h}(t)$ where

$$\begin{aligned}g(x) &= x^3, \quad h(x) = \frac{-\nabla \cdot (f_s p_s)}{p_s}, \quad p_s = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2} \\ \implies h(x) &= \frac{x}{\sigma^2} \\ \implies \chi(t) &= \frac{1}{\sigma^2} \mathbb{E}((X_t^0)^3 X_0^0)\end{aligned}$$

Use the fact that $X_t^0 = X_0^0 e^{-t/\tau} + J_t$ we have

$$\frac{1}{\sigma^2} \mathbb{E}((X_0^0 e^{-t/\tau} + J_t)^3 X_0^0) = \frac{1}{\sigma^2} \mathbb{E}((X_0^0)^4 e^{-3t/\tau} + 3J_t(X_0^0)^3 e^{-2t/\tau} + 3J_t^2(X_0^0)^2 e^{-t/\tau} + J_t^3 X_0^0)$$